



CHAPTER 1

Project Overview

1.1 Project Name

Multi-Platform Barangay Announcement and Garbage Collection Notifier using IOT Technology

1.2 Project Members

- Ambuyo, Joyce Bernadeth S., - **Documentation Leader**
- Canlas, Kenth Luigi B., - **Communication Leader**
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- Nadonga, Sunday Le'Chaim L., - **Project Leader**
- Nopre, Argie C., - **Technical Leader**
- Perdido, Hanna Llouise C., - **Quality Assurance Leader**

1.3 Project Duration and Activities

The Data Gathering, Planning, and Designing phase of the project will take place during the first semester of School Year 2025–2026, during this phase, the team will conduct requirements analysis, gather data through interviews or surveys, and create detailed plans for the project.

This will include defining the project's scope, creating system designs such as wireframes and database schemas, and outlining the overall architecture of the system.

The Development, Implementation, Testing, and Integration phase will be carried out during the first semester of School Year 2026–2027. In this phase, the actual coding and system development will take place, followed by implementation of the system. After development, the system will undergo rigorous testing, including unit testing, integration testing, and user acceptance testing. The final step in this phase will involve integrating the system into the intended environment and preparing documentation for final evaluation or presentation.



1.4 Background

In Barangay 133, Tondo, Manila, the team observed that communication between the barangay officials and the residents is a significant challenge. The current system for sharing information relies on traditional printed notices and posts on social media. However, these methods are often ineffective. Printed notices can be easily missed or ignored, while social media posts may not reach all residents due to algorithms or because not everyone follows the page. This leads to missed announcements, confusion, and delayed updates.

A more specific problem identified in the community is the inefficient waste management, particularly the "floating" garbage collection schedule. Residents are often unsure of the exact time the garbage truck will pass by. This uncertainty causes two main issues: first, residents leave their garbage out on the street for long hours, which attracts pests, creates bad smells, and poses sanitation and health risks. Second, many residents miss the collection entirely, leading to uncollected garbage piling up in the area.

The barangay currently lacks a centralized and real-time platform to handle these issues. There is no official system where residents can receive instant, verified alerts and where officials can efficiently manage services.

To address these problems, this study proposes the development of the **"Multi-Platform Barangay Announcement and Garbage Collection Notifier using IOT Technology."** This project aims to provide a reliable, technology-based solution to improve communication and streamline the garbage collection process, thereby helping Barangay 133 promote cleanliness, order, and better community participation.

1.5 Scope

The study is focused on the design, development, and evaluation of a complete, integrated system. The scope encompasses the following key features and functionalities:

1.5.1 Included

- User Account and Role Management
- Centralized Announcement System
- Image Processing-Based Garbage Collection Notifier through IP CCTV
- Residents Web Portal
- Feedback and Concerns Management System
- Centralized Database Backend



- Truck detection

1.5.2 Excluded

- Real Time GPS Tracking
- AI-Based Garbage Classification
- Financial Transaction or Payment Features
- Voice or SMS Notification Feature
- Integration with other Social Media Platforms



CHAPTER 2

Requirements Specification

This chapter presents a comprehensive overview of the requirements and specifications necessary for the successful development and implementation of the system. It outlines both functional and non-functional requirements, provides detailed use cases, and describes the technical, operational, and business aspects that guide the system design. By establishing a clear understanding of the system's expectations, constraints, and operational environment, this chapter serves as a blueprint for ensuring that the final product meets stakeholder needs and performs reliably under expected conditions.

2.1 Functional Requirements

Table 2.1 Functional Requirements

ID	Requirement Description	Priority
FR1	The system shall allow the Super Admin to add, update, and delete user accounts.	High
FR2	The system shall allow the Super Admin to add, update, and delete resident records.	High
FR3	The system shall allow Barangay Officials to log in using their assigned credentials.	High
FR4	The system shall allow Barangay Officials to create, update, and delete announcements.	High
FR5	The system shall allow Barangay Officials to view residents' history and feedback.	Medium
FR6	The system shall detect garbage trucks through CCTV camera input using image processing.	High
FR7	The system shall automatically send real-time notifications to residents upon garbage truck detection.	High
FR8	The system shall allow residents to log in using credentials provided by the barangay.	High



ID	Requirement Description	Priority
FR9	The system shall allow residents to view announcements and garbage collection notifications.	High
FR10	The system shall allow residents to fill out and send feedback forms to barangay officials.	Medium
FR11	The system shall allow the Super Admin to view all residents' feedback and activity history.	Medium
FR12	The system shall send push notifications to residents about new announcements.	Medium
FR13	The system shall store and retrieve all system data in a centralized database server.	High
FR14	The system shall allow officials to generate and view summary reports of announcements and resident activities.	Medium
FR15	The system shall display all information (announcements, alerts, feedback) in a web-based portal and mobile app accessible via desktop or mobile devices.	High
FR16	The system shall allow residents to view a log of their personal activity history.	Low



2.1.1 Functional Requirements Descriptions

Item	Description
Req ID	FR1
Description	The system shall allow the Super Admin to add, update, and delete both Barangay Official Admin and Resident accounts.
Priority	High
Functional Requirement Description	This requirement ensures that the Super Admin has full control over user management. It allows account creation, modification, and removal to maintain an accurate and secure list of system users.
Acceptance Criteria	1. Authentication and Authorization: Only the "Super Admin" has the privilege to the "User Management Page" 2. User Management Interface: The system provides an interface that displays a list of all existing user accounts, their roles, and their status. 3. Account Creation: • The interface must have an "Add User" button that opens a form.
	• The form must include fields for Username, Temporary Password, and User Role. • Validation must ensure the username is unique. • On success, the new user is added to the list, and a success message is shown. 4. Account Update: • The interface provides an "Edit" option for each user. • The Super Admin can modify the user's role or reset their password.



	● On success, the changes are saved and a success message is shown. 5. Account Deletion: ● The interface provides a "Delete" option for each user. ● The system must show a confirmation prompt before deleting. ● On confirmation, the user is removed or deleted.
Req ID	FR2
Description	The system shall allow the Super Admin to add, update, and delete resident records.
Priority	High
Functional Requirement Description	Allows the Super Admin to manage resident information for efficient waste collection, notification, and feedback services.
Acceptance Criteria	1. Authentication and Authorization: Only the "Super Admin" can access the Resident Records. 2. Resident Record Interface: The system provides an interface to view, search, and filter all resident records. 3. Resident Creation: ● The interface has an "Add Resident" button that opens a form. ● The form must include fields for resident details (e.g., Name, Address, Birthday, Contact Info). ● Validation ensures all required fields are filled.



	● On success, the new resident record is saved, and a success message is shown. 4. Validation (Duplicate Check): The system must prevent the creation of duplicate residents based on a unique combination (e.g., Full Name and Birthday). 5. Resident Update: ● The interface provides an "Edit" option for each resident. ● On success, the updated resident information is saved.
	6. Resident Deletion: ● The interface provides a "Delete" option for each resident. ● The system must show a confirmation prompt before deleting. ● On confirmation, the resident's record is removed or archived.
Req ID	FR3
Description	The system shall allow Barangay Officials to log in securely using authorized credentials.
Priority	High
Functional Requirement Description	This requirement enables Barangay Officials to access the admin dashboard through a secure login process. The system verifies credentials before granting access to prevent unauthorized entry.



Acceptance Criteria	1. Login Interface: The system provides a login page with fields for "Username" and "Password" and a "Login" button. 2. Authentication (Success): When a Barangay Official enters their correct credentials and clicks "Login," the system must grant them access and redirect them to their official dashboard. 3. Authentication (Failure): When a user enters an incorrect username or password, the system must display an error message (e.g., "Invalid username or password") and must not grant access. 4. Validation: If the "Username" or "Password" field is left empty, the system must show a message prompting the user to fill in the fields.
Req ID	FR4
Description	The system shall allow Barangay Officials to create, update, and delete barangay announcements.
Priority	High
Functional Requirement Description	This ensures that barangay officials can manage announcements efficiently. It provides the ability to post important updates such as meetings, garbage schedules, or emergency alerts.
Acceptance Criteria	1. Authentication and Authorization: Only logged-in "Barangay Official" can access the "Manage Announcements" page. 2. Announcement Interface: The system provides an interface to view all existing announcements.



	3. Announcement Creation: • The interface has a "Create Announcement" button that opens a form. • The form includes fields for "Title" and "Body/Content." • When saved, the new announcement is published and visible to residents. 4. Announcement Update: • The interface provides an "Edit" option for each announcement. • When saved, the changes are immediately reflected on the published announcement. 5. Announcement Deletion: • The interface provides a "Delete" option for each announcement. • The system must show a confirmation prompt before deleting. • On confirmation, the announcement is removed and no longer visible and archived.
Req ID	FR5
Description	The system shall allow Barangay Officials to view resident history and feedback.
Priority	Medium



Functional Requirement Description	Allows barangay officials to access residents submitted feedback and review their activity history for better community response and decision-making.
Acceptance Criteria	1. Authentication and Authorization: Only logged-in "Barangay Official" can access the "Resident History" and "Feedback" modules. 2. Data Access Interface: • The system provides a search function to find a specific resident. • When a resident is selected, the official can view a read-only log of their history (e.g., past requests, issues). • The official can view a read-only log of feedback submitted by that resident. 3. Data Integrity: This interface must be "view-only." The official must not be able to edit or delete any history or feedback logs from this screen.
Req ID	FR6
Description	The system shall detect garbage trucks through CCTV camera input using image processing.
Priority	High
Functional Requirement Description	Integrates image processing to automatically identify garbage truck presence at collection points using live CCTV footage.



Acceptance Criteria	1. System Connection: The system must successfully connect to and receive the live video feed from the designated CCTV cameras via RTSP. 2. CPU Optimization: The Python application must aggressively downscale the video stream to a 320x320 resolution before inference. 3. Frame Skipping: The system must implement a frame-skipping strategy (e.g., processing 1 frame every 10) to allow the CPU to keep up. 4. Detection, Validation, and Performance: The system's INT8-quantized TFLite model must demonstrate a proof-of-concept with a target accuracy of 85-90%. The system must maintain a low false-positive rate (no more than 1 false positive per hour) under daylight and clear-weather conditions. 5. Logging and Trigger: A confirmed detection must immediately trigger a secure HTTP API call to the main server (PC). The main server is then responsible for logging the event (FR13) and triggering the notification (FR7).
Req ID	FR7



Description	The system shall automatically send real-time notifications to residents upon garbage truck detection.
Priority	High
Functional Requirement Description	Sends automated notifications to residents when a garbage truck is detected near their collection area, ensuring timely waste disposal.
Acceptance Criteria	1. System Trigger: This function must be automatically triggered only by a successful detection event from FR6. 2. Notification Logic: ● The system must identify the correct group of residents to notify based on the camera location.
	● The system sends a notification to all targeted residents. ● The notification message must contain clear, predefined content (e.g., "The garbage truck is now in your area."). 3. Performance and Logging: ● Notifications must be sent within a specified time (e.g., 60 seconds) of the detection. ● The system must log all sent (or failed) notifications, including the timestamp and recipient group.
Req ID	FR8



Description	The system shall allow residents to log in using credentials provided by the barangay.
Priority	High
Functional Requirement Description	Provides residents with secure login credentials to access their personalized dashboards for announcements, notifications, and feedback.
Acceptance Criteria	1. Login Interface: The system must provide a login page with fields for "Username" and "Password" and a "Login" button for residents. 2. Authentication (Success): When a resident enters their correct, assigned credentials and clicks "Login," the system must grant them access and redirect them to the resident portal. 3. Authentication (Failure): When a resident enters an incorrect username or password, the system must display an error message (e.g., "Invalid username or password") and must not grant access. 4. Validation: If either the "Username" or "Password" field is left empty, the system must show a validation message prompting the user to fill in the fields.



Req ID	FR9
Description	The system shall allow residents to view announcements and garbage collection notifications.
Priority	High
Functional Requirement Description	Enables residents to access and view barangay announcements and automated garbage collection alerts through their user interface.
Acceptance Criteria	1. Authentication and Authorization: Only a logged-in resident can view the "Announcements" and "Notifications" pages. 2. Announcements Interface: The system must display a page or feed with a list of all published announcements from barangay officials,
	- ordered from newest to oldest. 3. Garbage Notifications Interface: The system must display a page or log of garbage collection notifications (from FR7), clearly showing the most recent or active notification. 4. Data Display: This interface must be "view-only." The resident cannot edit or delete any announcements or notifications.



Req ID	FR10
Description	The system shall allow residents to fill out and send feedback forms to barangay officials.
Priority	Medium
Functional Requirement Description	Provides residents a feedback submission feature for reporting issues, complaints, or suggestions directly to barangay officials.
Acceptance Criteria	1. Authentication and Authorization: Only a logged-in resident can access the "Send Feedback" feature. 2. Feedback Form Interface: The system must provide a form with fields for "Subject" (or category) and "Feedback Message." 3. Validation: The "Feedback Message" field must not be empty. 4. Submission: When the resident clicks "Submit," the feedback form is sent to the barangay official, and the resident receives a success confirmation message (e.g., "Feedback sent successfully."). 5. Duplicate Prevention: The system shall prevent a resident from submitting identical feedback (based on message content) within a 5-minute window to avoid spam.



Req ID	FR11
Description	The system shall allow the Super Admin to view all residents' feedback and activity history.
Priority	Medium
Functional Requirement Description	Allows the Super Admin to review residents' interactions, feedback, and activity records for performance monitoring and system improvements.
Acceptance Criteria	1. Authentication and Authorization: Only a user with "Super Admin" privileges can access the "Resident Activity" and "Feedback" logs. 2. Data Access Interface: The system provides a central dashboard to view all feedback submitted by residents and a log of all resident activities (e.g., logins, submissions).
	3. Search and Filter: The Super Admin must be able to search for a specific resident or filter the logs by date range or activity type. 4. Data Integrity: This interface must be "view-only." The Super Admin cannot edit or delete the logs.
Req ID	FR12



Description	The system shall send push notifications to residents about new announcements.
Priority	Medium
Functional Requirement Description	Automatically alerts residents when new barangay announcements are posted to keep them informed and updated.
Acceptance Criteria	1. System Trigger: This function must be automatically triggered when a "Barangay Official" publishes a new announcement (from FR4). 2. Notification Logic: • The system identifies the correct resident group to notify. • The system identifies the correct resident group to notify. Message content must be clear and predefined (e.g., "Barangay Announcement: [Announcement Title]"). 3. Performance and Logging: • Notifications must be sent within 60 seconds. • System logs all sent or failed notifications with timestamp and recipient details.
Req ID	FR13



Description	The system shall store and retrieve all system data in a centralized database server.
Priority	High
Functional Requirement Description	Ensures all data (user info, announcements, notifications, feedback) are stored in a centralized MySQL (or MariaDB) database server running on the on-premises PC.
Acceptance Criteria	1. Data Storage (Write): All data created (e.g., new user, resident record, announcement, feedback) must be successfully saved to the central MySQL database. 2. Data Retrieval (Read): All information displayed in the portal must be fetched from the central MySQL database.
	3. Data Persistence: Data must be persistent, meaning it is not lost after a user logs out or the server restarts.
Req ID	FR14



Description	The system shall allow officials to create and view summary reports of announcements and resident activities.
Priority	Low
Functional Requirement Description	This requirement enables officials to monitor system usage and resident engagement through summarized activity logs and reports.
Acceptance Criteria	1. Report Functionality: Officials must be able to access the "report" dashboard. 2. System Logic: Reports must accurately reflect data from the MySQL database. 3. Specific Reports: The system shall allow officials to generate a PDF report of all announcements posted within a selected date range.
Req ID	FR15



Description	The system shall display all information (announcements, alerts, feedback) in a web-based portal accessible via desktop or mobile devices.
Priority	Medium
Functional Requirement Description	Ensures the entire system is web-based and responsive using React (frontend) and a Python-based API (backend).
Acceptance Criteria	1. Accessibility: The system must be accessible using a standard web browser (e.g., Chrome, Safari, Firefox). 2. Responsive Design (Desktop): The web portal must be fully functional and display correctly on desktop and laptop screen resolutions. 3. Responsive Design (Mobile): The web portal's layout must automatically adapt to smaller screens (e.g., smartphones, tablets), ensuring all text is readable and all buttons are usable without requiring horizontal scrolling.
Req ID	FR16



Description	The system shall allow residents to view a log of their personal activity history.
Priority	Medium
Functional Requirement Description	This requirement enables residents to review their system interactions, such as login times, the status of their submitted feedback, and a log of all received notifications (both announcements and garbage alerts). This provides transparency and allows residents to track their engagement with the system.
Acceptance Criteria	1. Authentication and Authorization: Only a logged-in resident can access the "My Activity History" page. The resident must only be able to view their own history and not the activity of any other user. 2. Interface and Access: The Resident Portal must include a clearly labeled "Activity History" or "My Logs" option in the user menu/dashboard. 3. Data Display: The system must display a chronological, read-only list of the resident's past activities. This list must include, at minimum: Date and Time of Activity, Activity Type (e.g., Login, Feedback Sent, Notification Received), and a Status/Summary.



4. Sorting and Filtering: The system shall allow the resident to sort the history by date (newest first by default) and filter the history by Activity Type. * Data Integrity: This interface must be view-only. The resident must not have any interface options to edit, delete, or modify any logged activity records.

2.2 Non-Functional Requirements

ID	Non-Functional Requirement	Description
NFR1	Performance	The web application shall load all pages in under 3 seconds. Admin dashboards with complex data shall load in under 5seconds.
NFR2	Real-Time Processing	Garbage-truck detection (FR6) and resident notification (FR7)shall occur within 90 seconds of the event. This includes CPUdetection time on the Raspberry Pi plus the network API call tothe main server.
NFR3	Usability	The system shall be learnable. A new user must be able to log inand submit a feedback form in under 90 seconds.
NFR4	Reliability	The system's operation is dependent on the 24/7 operation ofboth the PC Server and the RPi Sensor, and the LAN connectionbetween them.



NFR5	Security	All web traffic must be encrypted in transit via the CloudflareTunnel (HTTPS). All user passwords stored in the database mustbe hashed (e.g., bcrypt).
NFR6	Maintainability	The codebase shall adhere to professional coding standards(e.g., docstrings, linting).
NFR7	Scalability	The system shall support at least 100 simultaneous active users.
NFR8	Availability	The system shall be accessible 24/7, assuming a stable powersupply and internet connection for both the PC Server and theRPi Sensor.
NFR9	Compatibility	The system shall work on modern browsers (Chrome, Edge, Firefox, Safari) and adapt to different screen sizes.
NFR10	Data Integrity	All database transactions must be validated to prevent duplicateor lost data.
NFR11	Backup and Recovery	The MySQL database shall perform automatic backups (e.g., ascheduled mysqldump) at least once per day.
NFR12	Accessibility	The web portal shall be usable by people with common visual and motor disabilities (WCAG 2.1 Level AA, keyboardnavigation).



NFR13	Audit Trail	The system shall log all major administrative actions (add, edit, delete) for accountability.
NFR14	Portability	The system has two components: (1) The Detector Service, portable to a Raspberry Pi OS (64-bit) environment. (2) The MainServer, portable to a Windows/Linux PC environment running MySQL, Python (API).
NFR15	Compliance	The system shall comply with the Data Privacy Act of 2012 (Republic Act No. 10173).

2.3 Use Cases

Use Case ID	Name	Description	Actors
UC1	Manage Accounts	Allows the Super Admin to create, update, and delete user accounts in the system.	Super Admin
UC2	Manage Resident Record	Enables the Super Admin to view and manage resident information.	Super Admin

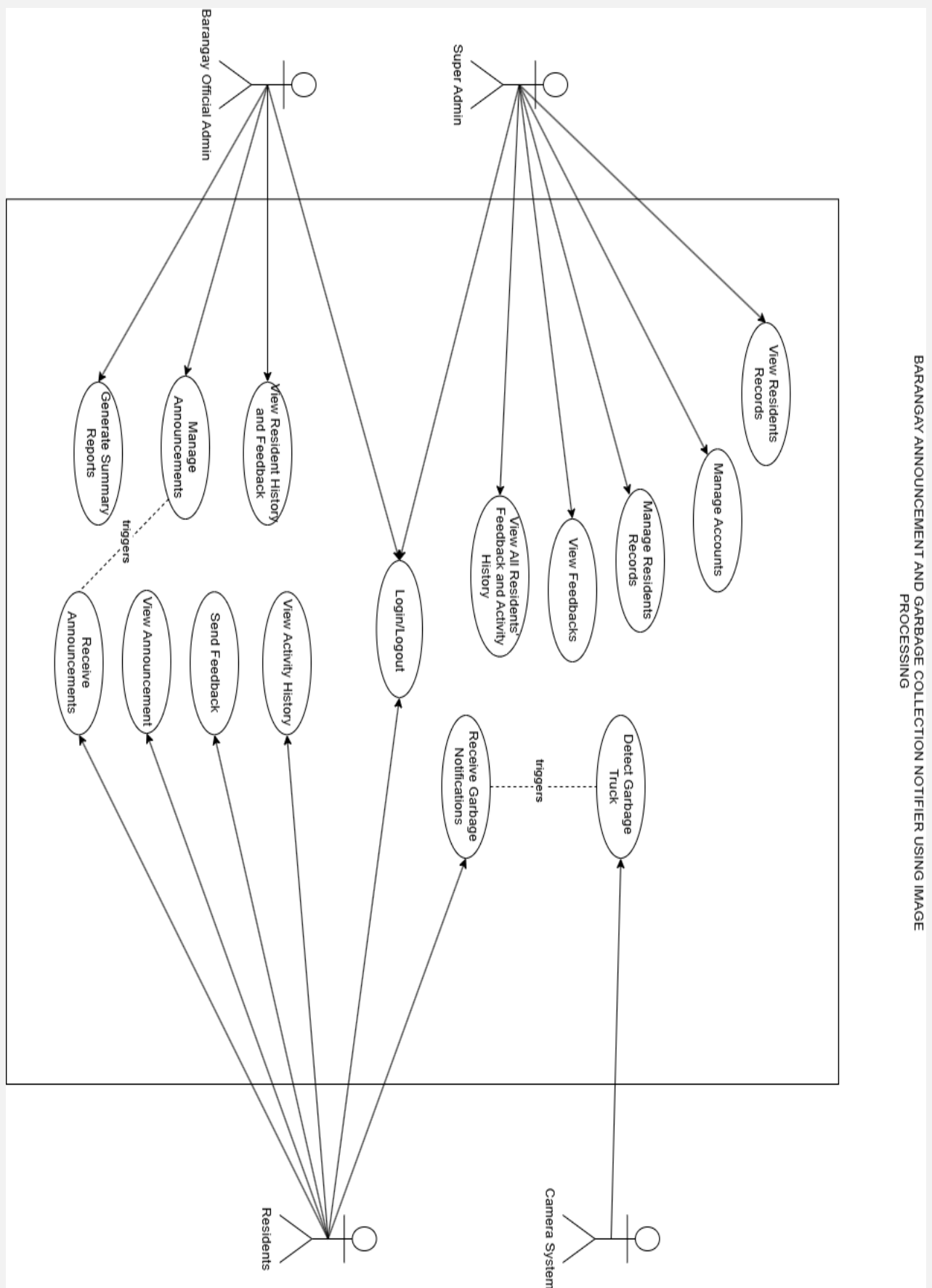


UC3	View Resident Feedback	Lets the Super Admin access and review feedback submitted by residents.	Super Admin
UC4	Manage Announcements	Allows the Barangay Official Admin to create, edit, and delete barangay announcements.	Barangay Official Admin
UC5	Send Feedback	Allows residents to send feedback or concerns to the barangay officials.	Residents
UC6	View Announcement	Lets residents view barangay announcements and garbage collection updates.	Residents
UC7	View Activity History	Enables residents to review their previous notifications and system activities as well as the feedback.	Residents



UC8	Login/Log out	Provides secure access to the system for all user types.	Super Admin, Barangay Official Admin, Residents
UC9	Receive Garbage Alert	Notifies residents automatically when a garbage truck is detected by the camera system.	Residents
UC10	Detect Garbage Truck	The system uses the live feed from the Camera System to identify a garbage truck in a predefined detection zone.	Camera System
UC11	Receive Announcement	Sends notifications to residents whenever new barangay announcements are posted.	Residents

2.3.1 Use Case Diagram





2.4 System Requirements

2.4.1 AI Training Environment (Cloud-Based)

This environment is used *only* for training and exporting the AI model. It is not used for the live operation of the system.

- **Service:** Google Colab
- **Purpose:** Training and fine-tuning the YOLOv8n object detection model on a custom dataset.
- **Framework:** Python, PyTorch / TensorFlow.

2.4.2 Deployment Server

This is the central machine that will run the AI model, backend, and database 24/7.

- **Device:** Raspberry Pi 5
- **Memory (RAM):** 4 GB
- **Storage:** 64GB+ High-Speed microSD Card (Class A2, U3) or External USB 3.0 SSD
- **Operating System:** Raspberry Pi OS (64-bit)

Software

- **AI/Backend:** Python 3.10+
 - **AI Model Runtime:** TensorFlow Lite (TFLite)
 - **AI Model:** YOLOv8n-TFLite (INT8 Quantized)
 - **Web API:** FastAPI or Flask
 - **Database:** MySQL/MariaDB
 - **Web Server:** CloudFlare Tunnel
 - **Frontend:** Web Portal (Admins & Officials): **React.js (hosted as static build files)** Mobile Application (Residents): **Flutter (Dart)**

2.4.3 Client-Side Requirements (User Devices)

- **For Barangay Official & Super Admin:** Any modern desktop or laptop.
- **For Residents:** Any smartphone, tablet, or computer.
- **Browser:** Latest version of Chrome, Safari, Firefox, or Edge.

2.4.4 Infrastructure & Network Requirements

- **CCTV System:** 1080p cameras with network streaming support (RTSP or ONVIF).



- **Processing Stream:** The **Raspberry Pi (2.4.2)** must downscale the 1080p RTSP feed to 320x320 for inference.
- **Processing Strategy:** The **Raspberry Pi (2.4.2)** must implement a frame-skipping technique.
- **Local Network (LAN):** A reliable **1 Gbps (Gigabit) Ethernet** network is required, connecting the CCTV DVR/NVR, the Raspberry Pi (2.4.2), and the Main Application Server (2.4.3).
- **Internet Connection:** 25 Mbps Symmetric (for Cloudflare Tunnel).
- **Secure Access & SSL:** The **Cloudflare Tunnel** service runs on the **Main Application Server (2.4.3)** to securely expose the web portal.

2.5 Business Rules

This section defines the business rules that govern the operation and management of the Barangay Announcement and Garbage Collection Notifier using Image Processing. These rules ensure the system operates consistently with barangay policies, maintains data privacy, and supports reliable automated notifications

1. **Account Creation and Management** – Only the Super Admin is authorized to create, update, and delete user and resident accounts. Residents are not permitted to self-register; credentials are issued by Super Admin to ensure identity verification and system security.
2. **Role-Based Access Control** – System access and privileges are determined by role: Super Admin (full system control), Barangay Official (manage announcements and feedback), and Resident (view announcements, receive notifications, submit feedback). Role-based access control enforces separation of duties and data protection.
3. **Announcement Posting** – Only authorized barangay officials or the Super Admin may create, edit, or delete announcements. Every announcement must include a title, message of body, and event date prior to publication.
4. **Announcement Visibility** – Announcements are displayed in chronological order with the most recent first. Each announcement shall remain visible for a minimum of seven (7) days unless removed by an authorized official.
5. **Feedback Submission** – Residents must be logged in to submit feedback. The system restricts duplicate submissions for the same issue to prevent spam. Barangay officials are expected to review and respond to feedback on a weekly basis.



6. **Image Processing Detection** – The system's image-processing module automatically detects the presence of a garbage truck using CCTV footage.
7. **Automatic Notification Trigger** – When the image-processing module confirms garbage truck detection, the system shall automatically send real-time notifications to all registered residents. No manual activation is required, ensuring a fully automated process for efficiency and reliability.
8. **Data Privacy and Security** – All personal and system data must comply with the Data Privacy Act of 2012 (Republic Act No. 10173). Sensitive information, including login credentials, shall be encrypted and accessible only by authorized personnel.
9. **System Availability** – The system shall operate continuously, 24 hours a day, seven days a week, except during scheduled maintenance. Barangay officials must announce any maintenance periods in advance.
10. **Detection Event Logging** – Each garbage truck detection event shall be automatically recorded with its timestamp, image reference, and notification status. This ensures traceability of every detection process.
11. **Ethical and Authorized Use** – All users must adhere to ethical use of the system. Tampering with data, posting false information, or misusing system features is strictly prohibited and subject to disciplinary action.

2.6. Constraints

1. Image processing accuracy may vary depending on lighting conditions, camera resolution, and angle of placement.
2. The detection feature functions only within predefined camera zones and cannot monitor unregistered areas.
3. Residents must be registered or logged in to receive announcements and notifications.
4. System performance may be affected by server capacity and network latency.
5. The project development and testing are limited to the 2nd semester of the academic year.
6. Hardware components such as cameras and servers must be available and properly maintained.



7. The system's accessibility depends on the compatibility of users' devices and browsers.

8. Maintenance and updates can only be performed when the server is online and stable.

9. The system, in its current scope, is dependent on user access to smartphones and the internet. It does not provide notifications for residents without digital access. 10. The system is a distributed system and depends on a reliable local network (LAN) connection between the Raspberry Pi 5 and the PC server.

11. The system requires two on-premises devices (the RPi 5 and the PC server) to be powered on and operational 24/7.

12. The AI detection (FR6) is still CPU-bound on the Raspberry Pi 5, limiting it to a low-FPS detection rate (1-2 FPS).

13. The system's public web accessibility is critically dependent on the Cloudflare Tunnel service (cloud flared) running on the PC server.

14. Image processing accuracy (FR6) may vary depending on lighting conditions, weather, camera resolution, and angle of placement.

2.7 Assumptions

1. Residents of Barangay 133 have access to smartphones or computers with stable internet connectivity.
2. Barangay officials and staff are trained to operate the admin dashboard and monitoring tools.
3. Surveillance cameras with sufficient resolution and reliability are already installed at designated garbage collection points.
4. Residents are willing to participate in the feedback system and keep their contact information updated for notifications.
5. The Barangay Chairman will assign one (1) personnel to manage all system-related data and oversee daily operations.
6. The database server and web platform are hosted locally or on a secure cloud service to ensure data accessibility and reliability.



7. The image processing system can be trained to accurately detect and identify garbage trucks under daylight and clear weather conditions.
8. Residents have basic digital literacy to navigate the website or mobile interface for announcements and feedback.

2.8 Dependencies

1. The system depends on the proper setup and operation of the camera module for image detection.
2. The barangay and residents have reliable internet connectivity.
3. The database must remain active to support announcements, feedback, and user records.
4. User cooperation is essential during system testing and deployment.
5. The system relies on a properly configured Python environment (Flask/Django) for backend operations.
6. Push notifications depend on third-party services and active network connections.
7. Secure access requires valid SSL certificates for encrypted data transmission.
8. Continuous operation depends on stable power supply and backup systems.
9. The system requires compatible web browsers and updated devices for optimal performance.
10. The system's detection accuracy is critically dependent on the quality, resolution (min. 1080p), and network accessibility of the existing barangay CCTV cameras. Performance with lower-specification cameras is not guaranteed.
11. The notification system (FR7) is critically dependent on a successful HTTP API call from the RPi 5 to the PC server.
12. The system's detection performance is critically dependent on the successful conversion of the YOLO model to an INT8-quantized TFLite file.
13. The system's public web access depends entirely on the cloudflare tunnel service running and authenticated on the Main Application Server (PC).
14. The system requires a registered domain name pointed to Cloudflare's nameservers.
15. The system depends on the proper setup and operation of the existing CCTV cameras and their RTSP streams.



16. Continuous operation depends on stable power supply and active cooling for the Raspberry Pi 5.
17. Continuous operation depends on a stable power supply for the PC Server.

CHAPTER 3

3.1 Architecture Diagram

The system follows a **Client-Server Architecture** structured within a **Four-Tier Layer Model**. In this architecture, the **Clients** (Residents' smartphones, Admin laptops, and the Raspberry Pi detection unit) initiate requests and display information, while the centralized **Server** (the Main PC) processes logic, manages data, and secures the database. This separation ensures that heavy processing and data storage remain secure on the server, while the user interface remains lightweight and responsive on client devices.

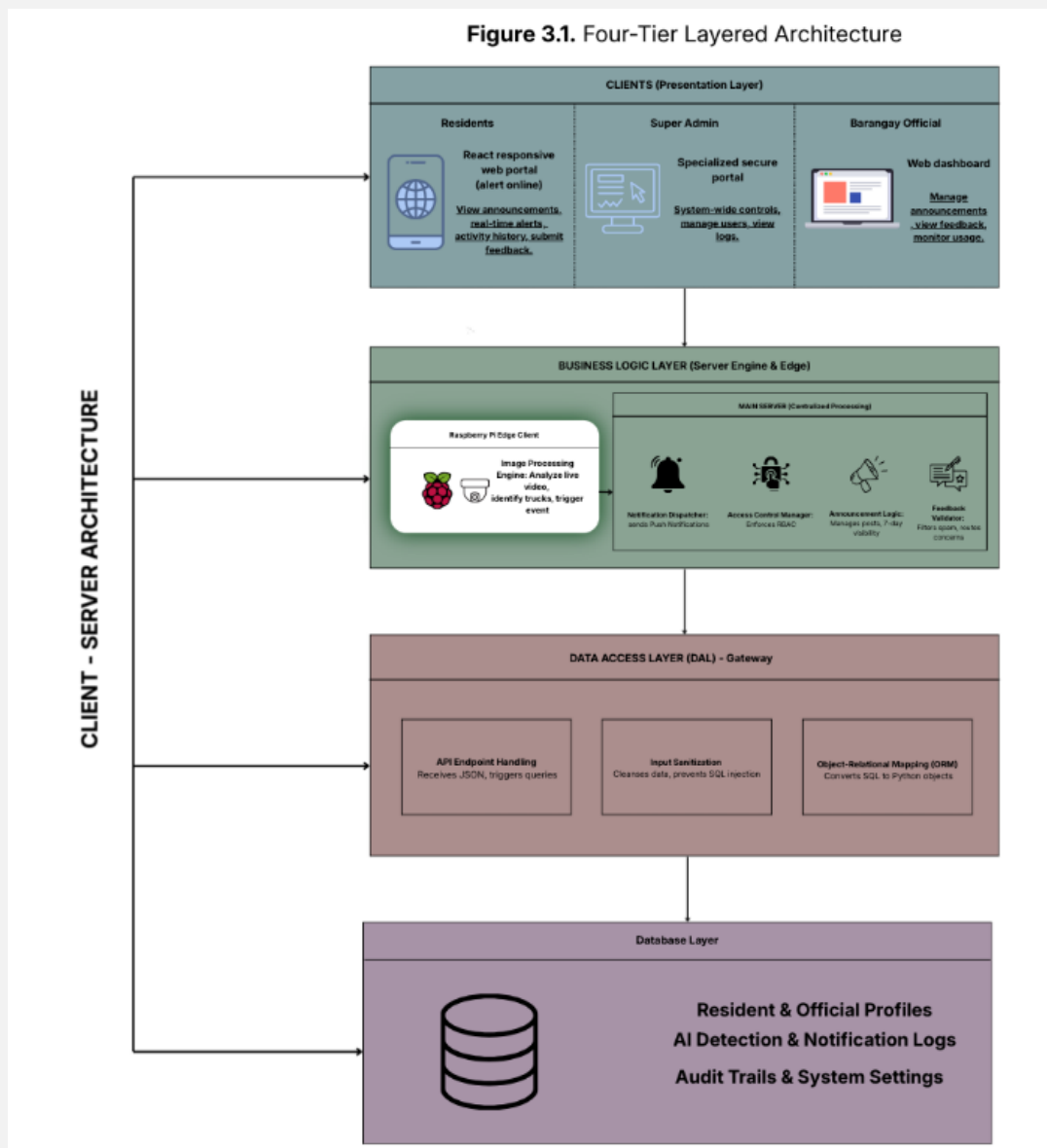


Figure 3.1. Four-Tier Layered Architecture



3.1.1. Presentation Layer: The User Interface

This layer represents the "Client" side of the architecture. It is responsible for gathering user input and presenting data fetched from the server.

Table 3.1. User, Interface Descriptions, and Context

User	Interface Descriptions	Context
Resident	Responsive web portal (React) displaying the "Garbage Notifier" alert, Dashboard, and Feedback Form.	Allows the residents to log in, view announcements, see realtime garbage collection alerts, view activity history, view and submit feedback or concerns to the barangay.
Barangay Official	Web dashboard with secure forms for creating announcements and viewing schedules.	Allows the official to create, edit, and delete announcements; view all resident feedbacks and activity history; and monitor system usage.
Super Admin	Specialized secure web portal with system-wide controls (User Management, Logs).	Allows the Super Admin to manage all user accounts, view system logs and reports, and oversee all data in the system.

3.1.2. Business Logic Layer: The Server Engine

This layer resides primarily on the **Server** (and partially on the Raspberry Pi Edge Client). It acts as the bridge, processing commands from the clients before they reach the data.



Table 3.2. Core Logic Component, System Action, and Proposal Benefit

Core Logic Component	System Action	Proposal Benefit
Image Processing Engine	Analyzes live video feeds using algorithms to identify garbage trucks and triggers an event upon detection	Automation: Solves the problem of "floating" schedules by automatically identifying truck arrivals without human intervention.
Notification Dispatcher	The Server receives the signal and instantly broadcasts a "Push Notification" to all connected Residents.	Timeliness: Ensures residents are notified instantly, preventing uncollected garbage and missed community updates
Access Control Manager	Enforces Role-Based Access Control (RBAC), ensuring strict separation between Super Admins, Officials, and Residents.	Security: Prevents unauthorized access to sensitive modules like the Admin Dashboard and User Management.
Announcement Logic	Manages the creation and visibility of posts, ensuring announcements remain visible for the mandatory 7-day period.	Reliability: Replaces ineffective printed notices with a centralized, verified source of information.



Feedback Validator	Processes resident concerns, filters out spam (duplicate submissions within 5 minutes), and routes valid feedback to officials.	Efficiency: Organizes community complaints into a formal system, eliminating the clutter previously experienced on social media.
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3.1.3. Data Access Layer (DAL): The Data Gateway

The DAL is a server-side component that abstracts the database. It ensures that Clients never communicate directly with the Database, preventing security vulnerabilities.

Table 3.3. Data Access Layer Functions, Technical Action, and Proposal Benefit

DAL Function	Technical Action	Proposal Benefit
API Endpoint Handling	Receives JSON requests from the React Client (e.g., GET /announcements) and triggers the appropriate database query.	Decoupling: Allows the Client (Frontend) and Server (Backend) to be developed independently.
Input Sanitization	Cleans all data sent by clients (e.g., removing special characters from feedback forms).	Protection: Prevents SQL Injection attacks where a malicious client tries to destroy the database.
Object-Relational Mapping (ORM)	Converts complex SQL table rows into simple Python objects that the Business Logic layer can understand.	Maintainability: Makes the server code cleaner and easier to update if database rules change.



3.1.4 Database Layer

This is the secure repository for all transactional and historical data. It is physically and logically protected from external access.

Table 3.4. Data stored, access requirement and proposal benefit

Data Stored	Access Requirement	Proposal Benefit
Resident & Official Profiles	Hashed Passwords (bcrypt); Role-Based Access Control.	Compliance: Strictly adheres to the Data Privacy Act of 2012 (RA 10173) by protecting sensitive resident information from unauthorized access.
AI Detection & Notification Logs	High Performance: Optimized for high-speed write operations (INSERTs) from the Raspberry Pi.	Real-Time Processing: Supports NFR2 by ensuring that rapid garbage detection events are recorded instantly without slowing down the notification engine.
Audit Trails & System Settings	Immutable Logging: Append-only access; restricted strictly to Super Admin.	Accountability: Fulfills NFR13 by maintaining a permanent, unalterable history of all administrative actions (login, create, delete) for security audits.



3.2 Module Descriptions

Table 3.5. Module Description

Module Name	Description	Example Functionality
1. Authentication & Security Module	The gatekeeper of the system. It handles secure login, session management, and password hashing (bcrypt) for all three user roles (Super Admin, Official, Resident).	Verifies that a user logging in is a "Barangay Official" and grants them access to the Admin Dashboard while denying them access to Super Admin settings.
2. Vision-Trak AI Detector (Edge)	The dedicated software running on the Raspberry Pi 5 . It captures live video from the CCTV and runs the YOLOv8 TFLite model to detect garbage trucks.	Processes a video frame, detects a "Garbage Truck" with 90% confidence, and immediately sends a "Positive Detection" signal to the main server.
3. Detection	The backend listener (Python API) that receives signals from the Raspberry	Receives a detection signal but ignores it if a "Truck Alert" was already sent to residents in the last 5 minutes (Spam Prevention).



Event Handler (Server)	Pi. It validates the signal to prevent false positives or duplicate spam alerts.	
4. Notification Broadcast Service	Manages the real-time dissemination of alerts. It pushes data to the React frontend to trigger the visual "Garbage Truck Arriving" banner on resident devices.	Instantly updates the status on the Resident Dashboard from "No Schedule" to "TRUCK DETECTED" when the signal is verified.
5. Announcement Management Module	Allows Barangay Officials to create, update, and delete community announcements. It manages the visibility and expiration of posts.modules whenever the truck is detected within the monitored area.	An official drafts a post titled "Barangay General Assembly," sets the date, and publishes it for all residents to see.
6. News Feed &	The front-facing React component that	A resident opens the website and sees the latest announcements at the top, followed by the current garbage collection status.



Dashboard Module	fetches and displays active announcements and truck status in chronological order for residents.	
7. Resident Feedback Collector	Provides the interface and validation logic for residents to submit complaints or suggestions regarding barangay services.	A resident fills out a form reporting "Uncollected Trash in Zone 4." The module ensures the message isn't empty before saving it.
8. Feedback Review & Response	An administrative module that allows Officials and Super Admins to view resident feedback, filter by date/topic, and mark issues as "Resolved."	A Barangay Official views a list of new complaints, filters them to find garbage-related issues, and marks a resolved complaint as "Done."
9. Resident Record Manager	A restricted module for the Super Admin to manage the database of residents. It handles the creation and	The Super Admin adds a new user to the system, inputting their names and assigning them a " Resident " role credential.



	updating of resident profiles.	
10. Official Account Controller	A highly restricted Super Admin module for creating and managing Barangay Official accounts.	The Super Admin creates a login account for a Barangay Official, giving them "Official" level permissions.
11. System Activity Logger	Automatically records all major actions (logins, deletions, edits) into an immutable log for security auditing and accountability.	Records a timestamped log entry: <i>"Admin_User1 deleted Announcement #45 on Nov 30 at 10:00 AM,"</i> ensuring actions can be traced.
12. Report Generation Engine	Aggregates system data to generate printable summaries (PDF/CSV) regarding detection frequency and user engagement.	Generates a monthly report showing: <i>"Total Trucks Detected: 24"</i> and <i>"Total Complaints Received: 5"</i> for the Barangay Captain's review.
13. Resident Interaction History Module	Ensures accountability by maintaining a permanent, searchable history of	A resident checks the "Announcement Archive" to prove that the Barangay Official posted a "No Collection" notice last Tuesday.



	all barangay announcements. Even "expired" or "deleted" posts are archived, so residents can verify past information.	
14. System Maintenance & Configuration	Manages database backups, monitors the Raspberry Pi connection status, and handles data cleanup to ensure the system runs 24/7 without crashing.	The Super Admin checks the "Server Health" status to see if the Camera is online and clicks "Backup Data" to save a copy of the records.

3.3 Data Design

The Data Design section is critical; it establishes the formal framework for how the system's information will be structured, managed, and processed, translating conceptual needs into a concrete blueprint that ensures data integrity, consistency, and efficient retrieval throughout the entire system lifecycle.



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provides a structured channel for residents to report issues to administrators. Furthermore, system integrity and administrative oversight are secured through tbl_AuditLogs, which records transactional history for security purposes, and tbl_Reports, which aggregates data from feedback and detection modules to assist officials in analyzing community cleanliness and system usage trends.

3.3.2 Data Flow Diagram

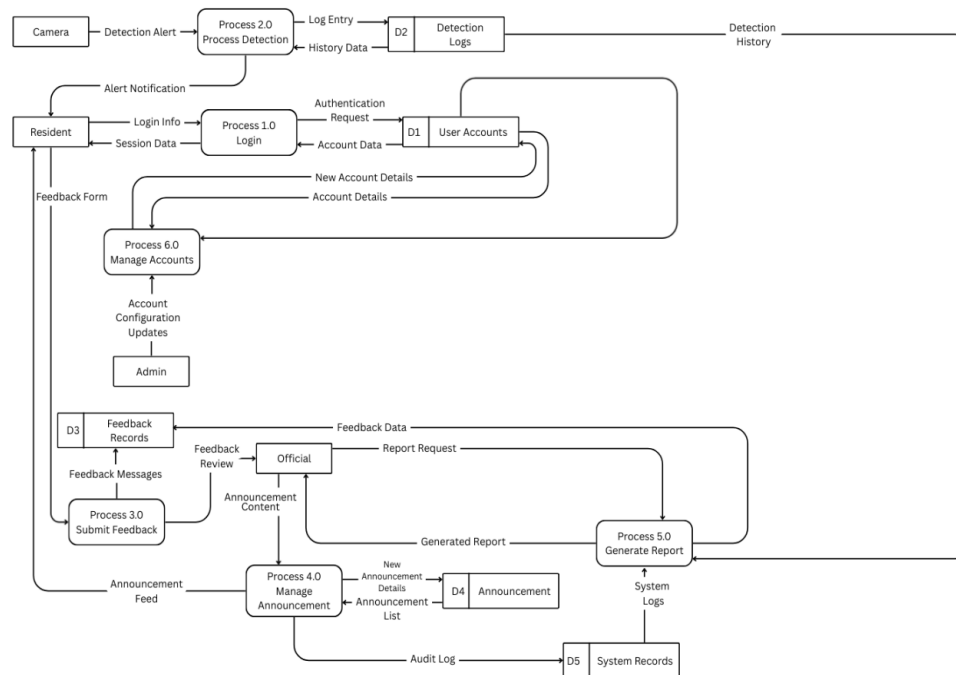


Figure 3.3. Data Flow Diagram

Figure 3.3 presents the Data Flow Diagram, illustrating the logical movement of data between external entities, processes, and data stores. The system interacts with three main external entities: Resident, Official, and Admin. Functionality is divided into six core processes. The Login Process handles Login Info from the Resident, verifies the account against the User Accounts data store, and allows the Resident to View Dashboard. The Process Detection receives Garbage Truck Detection signals from the Camera, checks the status against Detection Logs, and outputs Verified Data. The Submit Feedback Process receives Sends Feedback from the Resident, stores the Feedback Messages in Feedback Records, and allows the Resident to Review Feedback. The Manage Announcement Process receives the Creates



Announcement action from the Official, stores Announcement Details in the Announcement data store, and Publishes Announcement to the Resident. For administration, the Generate Report Process receives a Report Request from the Official, retrieves the Date and Time Range from System Records, and sends the PDF Summary back to the Official. Finally, the Manage Accounts Process receives requests from the admin to Add Residents and perform Config Update, manages Reset/Delete Accounts and Security History, and sends Audit Logs to the Admin. The DFD comprehensively models the flow of information across the system's core functions, from device input to administrative logging and service delivery.

3.3.3. Activity Diagram

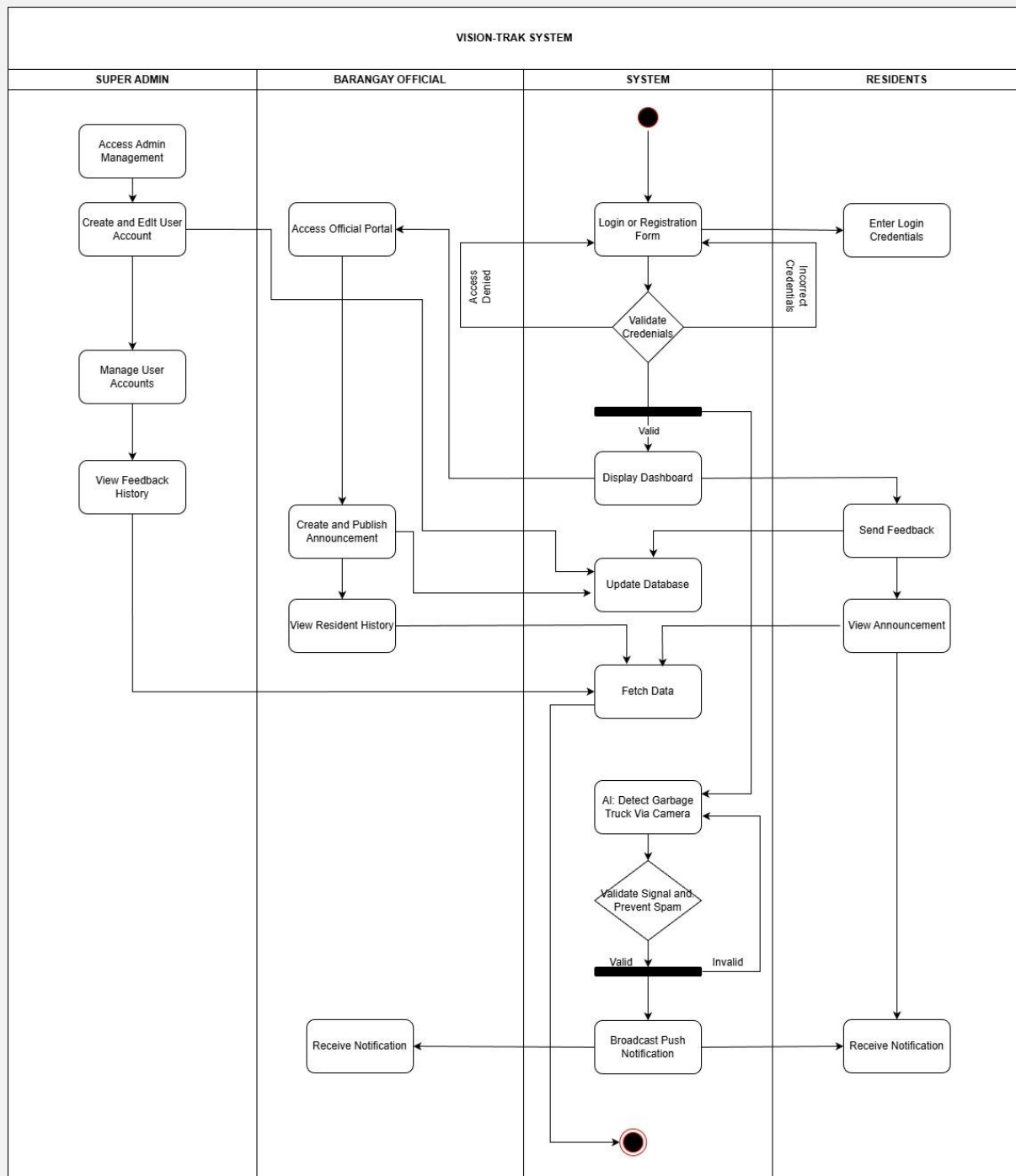


Figure 3.4. Activity Diagram

Figure 3.4 illustrates the Activity Diagram for the Vision-Trak system, modeling the functional workflows across four swim lanes representing the main actors: Super Admin, Barangay Official, Barangay System (the core application), and Resident. The diagram details the processes from user authentication to administrative tasks and the automatic garbage detection service. The process begins with the Login activity



initiated by the Super Admin, Barangay Official, or Resident. All login attempts proceed to the central system activity: Validate Credentials. If the validation fails for any user, the flow exits with an Invalid Credentials message directed back to the corresponding user type. If validation is successful, the flow synchronizes and leads to the Display Dashboard activity, granting role-based access. After accessing the dashboard, each user role gains parallel access to specific functionalities. The Super Admin can Manage Accounts and View Feedback and History. The Barangay Official can Manage Announcements and View Residents' History. The Resident can View Announcement and Send Feedback. Crucially, the administrative and official activities often interact with the core system. The Manage Accounts, Manage Announcements, and View Residents History activities all interact with the system's Update Database and Fetch Data activities, ensuring data consistency. The Super Admin, Barangay Official, and Resident all have independent Logout paths, terminating their respective sessions. A separate, automated flow exists within the Barangay System Lane, commencing with the independent service that Detects Garbage Truck. Upon detection, the system Sends Alert, which leads directly to the Resident activity Receive Notification, concluding the core real-time service mechanism. The entire diagram provides a comprehensive, sequential view of system operation, clearly distinguishing between user-initiated tasks and automated services.

3.3.4 Database Schema

Table 3.6. List of tables and Attributes

Table Name	Attributes	Description
tbl_Users	user_id (PK), username, password, roles	The central authentication table used to store login credentials and define the access level (role)



		for every user in the system.
tbl_Admin	admin_id (PK), user_id (FK), first_name, middle_name, last_name, birthday, gender, contact	Stores specific profile information for system administrators who have full control over system settings.
tbl_Official	official_id (PK), user_id (FK), first_name, middle_name, last_name, birthday, gender, contact	Contains the demographic details of community officials responsible for managing reports and announcements.
tbl_Residents	resident_id (PK), user_id (FK), first_name, middle_name, last_name, birthday, gender, address, contact	Stores the profile data of the community residents who use the system to view alerts and send feedback.
tbl_DetectionLog	log_id (PK), timestamp, confidence_score,	The primary operational table that records



	image_path, notification_status	automated detection events, including the AI confidence score and the file path to the captured evidence image.
tbl_Notifications	notification_id (PK), log_id (FK), status, sent_at	Manages the queue and status of alerts generated from detection logs to ensure relevant parties are informed.
tbl_Announcement	announcement_id (PK), created_by (FK), title, content, date_posted	specific record of broadcast messages posted by officials for the entire community to view.
tbl_Feedback	feedback_id (PK), created_by (FK), subject, content, timestamp	Stores complaints, suggestions, or concerns submitted by residents for review by officials and admins.



tbl_Reports	report_id (PK), title, file_format, report_type, start_date, end_date	Archives generated system summaries, categorized by type (e.g., garbage alerts, activity history) for administrative analysis.
tbl_SystemSettings	system_id (PK), updated_by (FK), config_key, config_value	Contains global configuration variables that control the system's behavior, accessible only by administrators.
tbl_AuditLogs	log_id (PK), user_id (FK), action_type, timestamp	A security table that maintains a chronological record of user activities to ensure accountability and traceability.

Table 3.6 presents the database schema designed to support the system's core functionality. The architecture is centered on the `tbl_Users` entity, which facilitates centralized authentication and Role-Based Access Control (RBAC) by linking to specialized profile tables: `tbl_Admin`, `tbl_Official`, and `tbl_Residents`. The system's automated monitoring capabilities are managed through `tbl_DetectionLog`, which stores machine learning outputs such as confidence scores and evidence images, while `tbl_Notifications` ensures the timely dissemination of alerts triggered by these logs. Community engagement is structured through

tbl_Announcement for official broadcasts and tbl_Feedback for resident concerns. Furthermore, administrative oversight and data integrity are maintained by tbl_SystemSettings for configuration management, tbl_Reports for data aggregation, and tbl_AuditLogs, which provides a verifiable trail of all user activities for security purposes.

3.5. Physical Arrangement of Devices

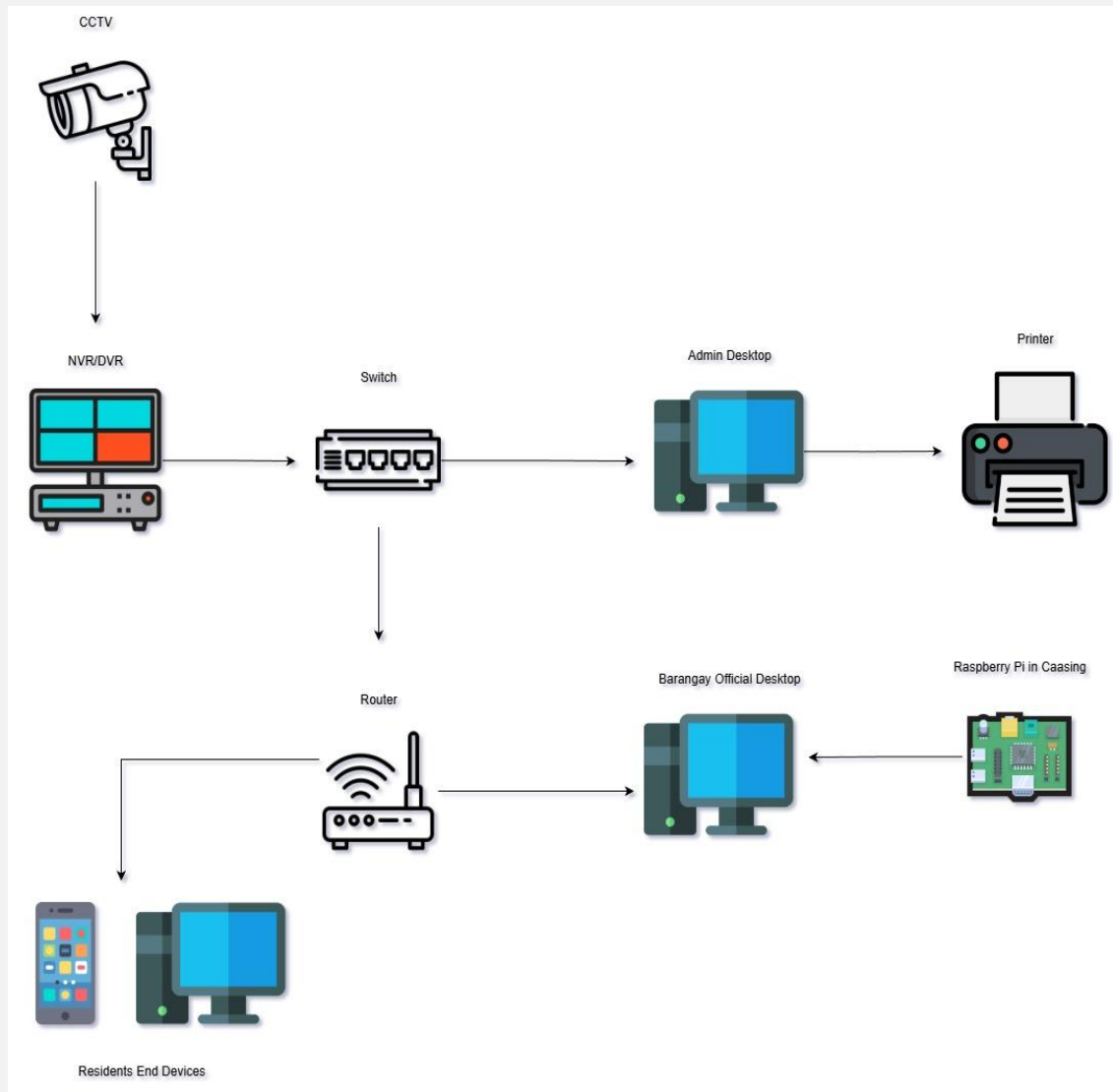


Figure 3.5. Hybrid network

Figure 3.5 illustrates the Hybrid Network architecture for the Barangay one three system, detailing the interconnected physical components, data flows, and network infrastructure. The diagram displays a combination of CCTV surveillance infrastructure and a standard Local Area Network structure. The surveillance system



begins with a CCTV camera, which is connected to storage and processing units. The video signal from the camera is channeled to a NVR (Network Video Recorder) and a DVR (Digital Video Recorder), with an HVR (Hybrid Video Recorder) positioned centrally to manage these streams. The video output from these recorders is displayed on a central CCTV Monitor and can also be viewed by dedicated computer terminals. The core data network provides connectivity for local processing and internet access. The CCTV system connects to a Switch, which acts as the central distribution point for the wired network. A Raspberry Pi device, designated for the Database and AI processing, is connected to the switch and feeds data to a primary workstation, PC two. The Switch also connects to the Router, which provides wireless access to Mobile Devices. Another workstation, PC one, is connected downstream and manages a printer and other peripherals, representing general administrative or management functions. This hybrid network design successfully integrates the crucial real-time video surveillance system with the administrative data network, ensuring both monitoring and data processing occur within a connected environment.

3.6. Conceptual Design

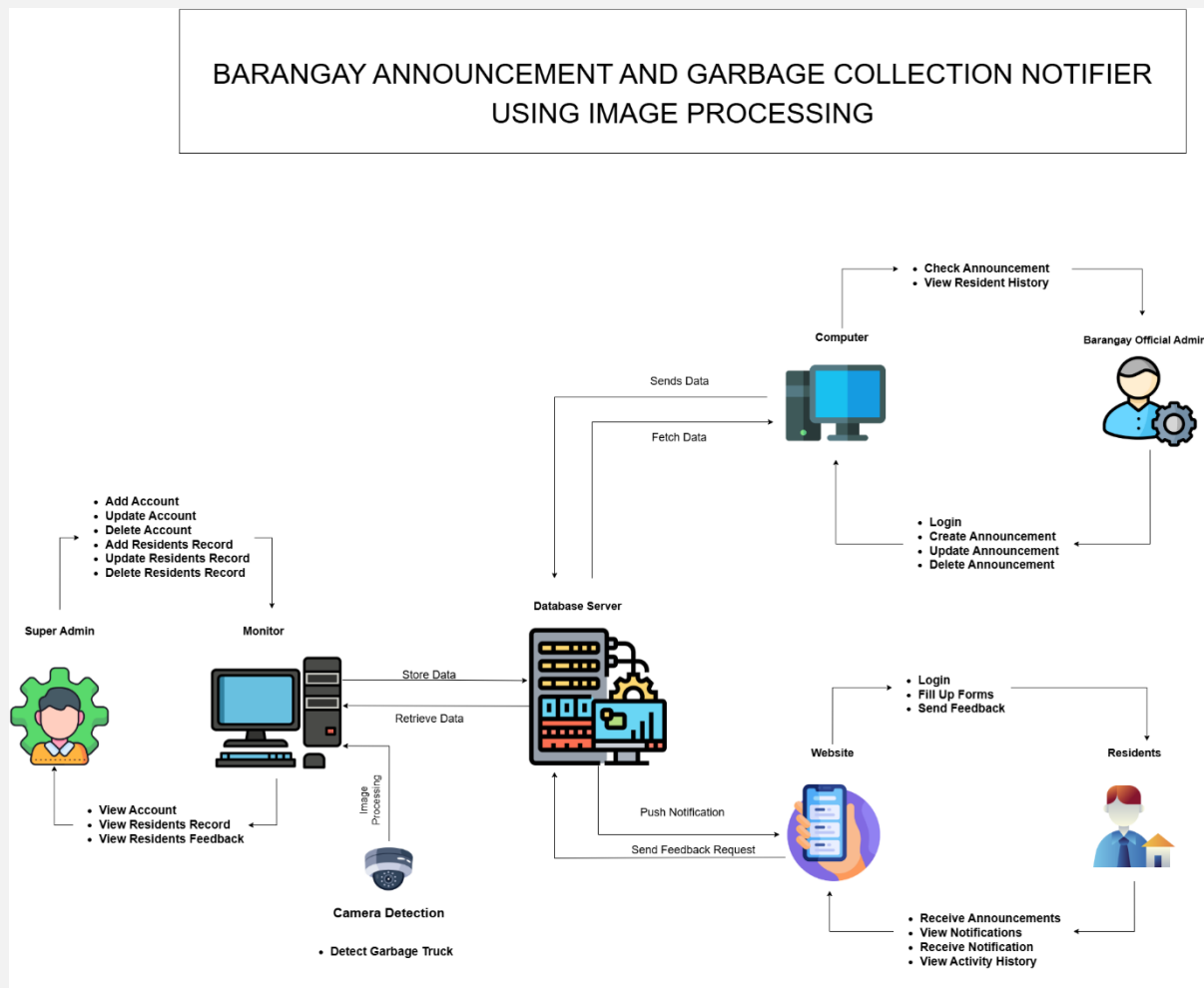


Figure 3.6. Proposed Conceptual Design

Figure 3.6 illustrates the Proposed Conceptual Design of the Barangay Announcement and Garbage Collection Notifier Using Image Processing, depicting the structural interplay between hardware components, user interfaces, and the central data repository. The architecture is anchored by a Database Server that facilitates real-time data synchronization across three primary operational modules. On the administrative side, the Super Admin is responsible for maintaining system integrity by managing user accounts and resident records, while simultaneously overseeing the automated monitoring component where a camera utilizes image processing to detect garbage trucks and store the resulting data. Parallel to this, Barangay Official Admins access the system via computer terminals to generate announcements and review resident history. The workflow extends to the community through a Resident Interface

(accessible via website or mobile), which supports bidirectional communication; this allows residents to receive instant push notifications regarding garbage collection and official announcements, while also providing a streamlined channel to submit feedback forms and view their personal activity history.

3.7. System Prototype

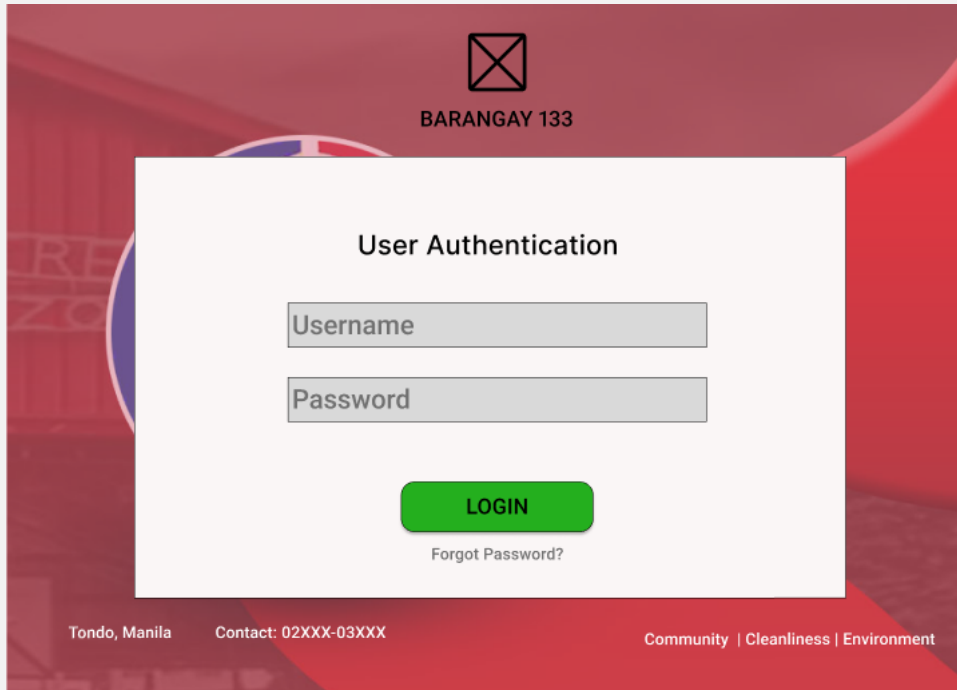


Figure 3.7.1. Login Form

Figure 3.7.1, the Login Form serves as the initial gateway for all users accessing the Barangay 133 system. It presents a clean and professional interface with a centered User Authentication panel, designed to accommodate existing users logging into their accounts. The form provides dedicated input fields labeled Username and Password, where users may enter their registered Barangay ID or Email Address and corresponding password to ensure secure authentication. A clearly visible LOGIN button is positioned below the input fields, functioning as the primary call-to-action for submitting user credentials. Additional support features are included to enhance usability, such as the “Forgot Password?” link located beneath the login button, which assists users in account recovery. Overall, the login form is designed to deliver a secure, user-friendly, and efficient access point consistent with the barangay’s official interface layout.

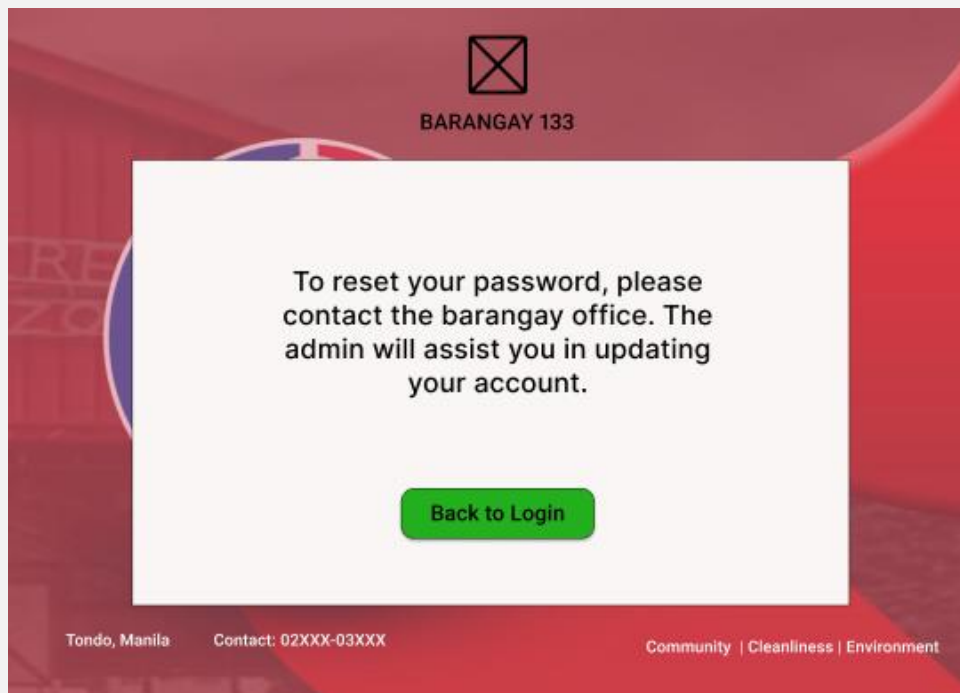


Figure 3.7.2. Forgot Password

Figure 3.7.2, the **Password Reset Interface** provides guidance for all users by instructing them to contact the barangay office for account recovery. A **"Back to Login"** button allows users to return to the login screen, ensuring a clear and secure password reset process.

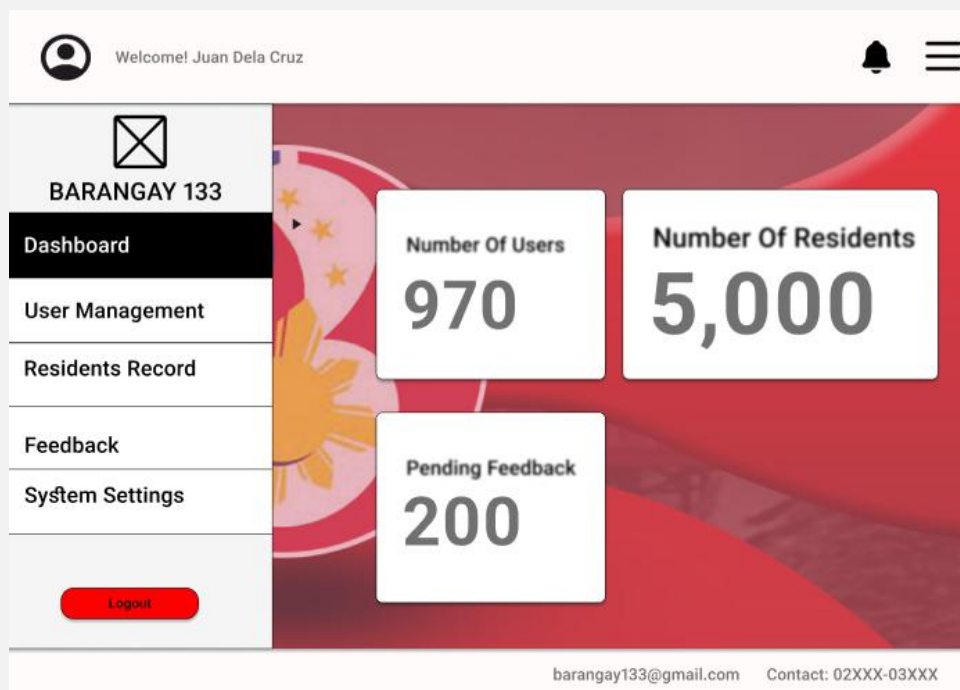


Figure 3.7.3. Admin Dashboard

Figure 3.7.3, presents the Admin Dashboard of the Barangay 133 system, designed to provide supervisory personnel with an immediate, high-level overview of system status and primary navigational control. The interface is structurally organized into a vertical Navigation Panel and the Content Area. The Navigation Panel features the currently highlighted Dashboard view, along with links for User Management, Residents Record, Feedback, System Settings, and a Logout control. The Content Area displays three crucial Key Performance Indicators: the Number of Users, the Number of Residents, and Pending Feedback. This interface functions as the central operational hub, enabling administrators to efficiently monitor system scale, prioritize actionable items, and access all system modules for streamlined management.

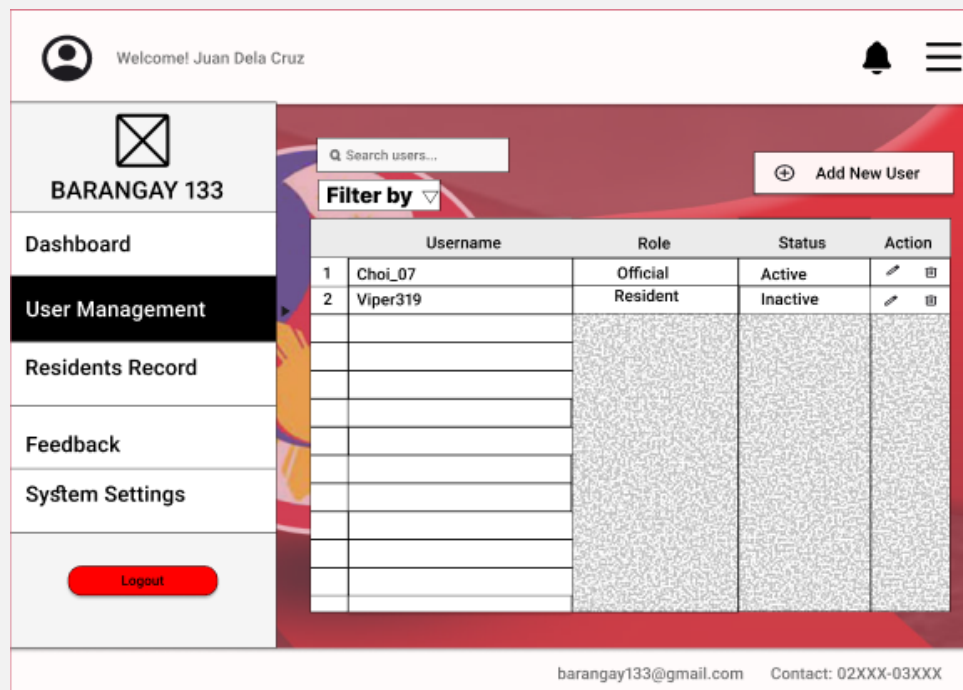


Figure 3.7.4. User Management

Figure 3.7.4 illustrates the User Management interface prototype, a critical administrative module for the Barangay 133 system designed to control system access and credentials. The view maintains the standard layout, with the Navigation Panel clearly highlighting the active User Management module. The Content Area is dedicated to account administration, featuring a Search Bar to locate specific users

and a Filter by Role control for list segmentation. Account provisioning is facilitated by the Add New User button. The central feature is a detailed table listing user records with defined columns for Username, assigned Role (e.g., Official, Resident), current Status (Active or Inactive), and Action controls (Edit and Delete). This module allows administrators to efficiently review, search, and manage all system user accounts and permissions.

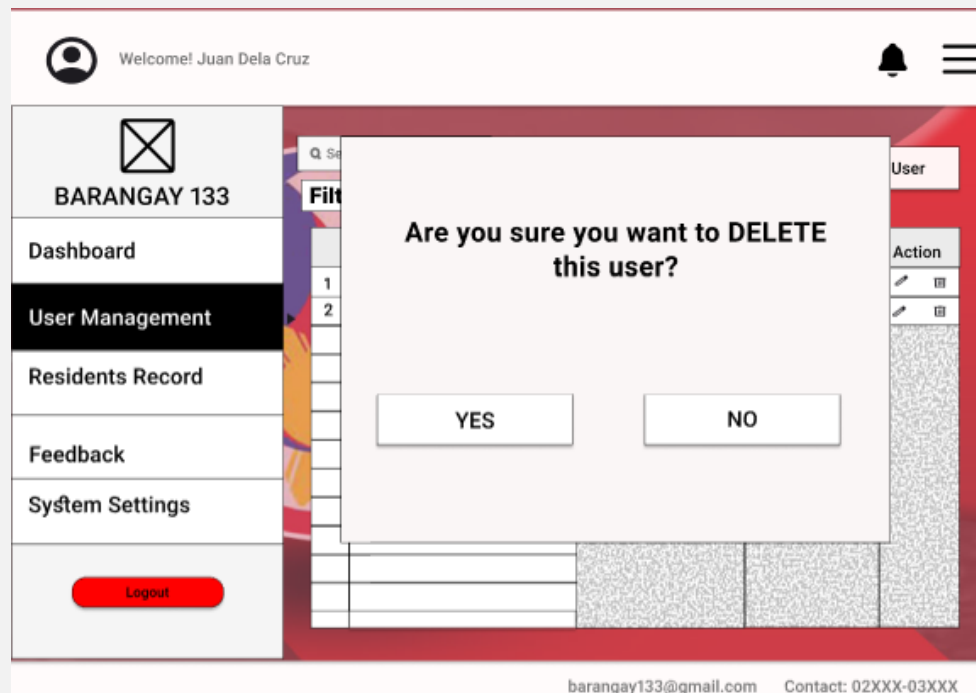


Figure 3.7.5. Delete Confirmation Dialog

Figure 3.7.5 shows a delete Confirmation Dialog, which appears as an overlay on the User Management interface prototype. This is a crucial system safeguard implemented to prevent accidental data loss. The dialog intentionally interrupts the administrative workflow, presenting the explicit verification prompt: "Are you sure you want to DELETE this user?" The administrator is required to provide a definitive response via two high-contrast options: "YES" to irrevocably proceed with the account deletion, or "NO" to safely abort the operation and remain on the User Management screen. This mechanism enforces necessary authorization and reduces the risk associated with irreversible administrative actions.

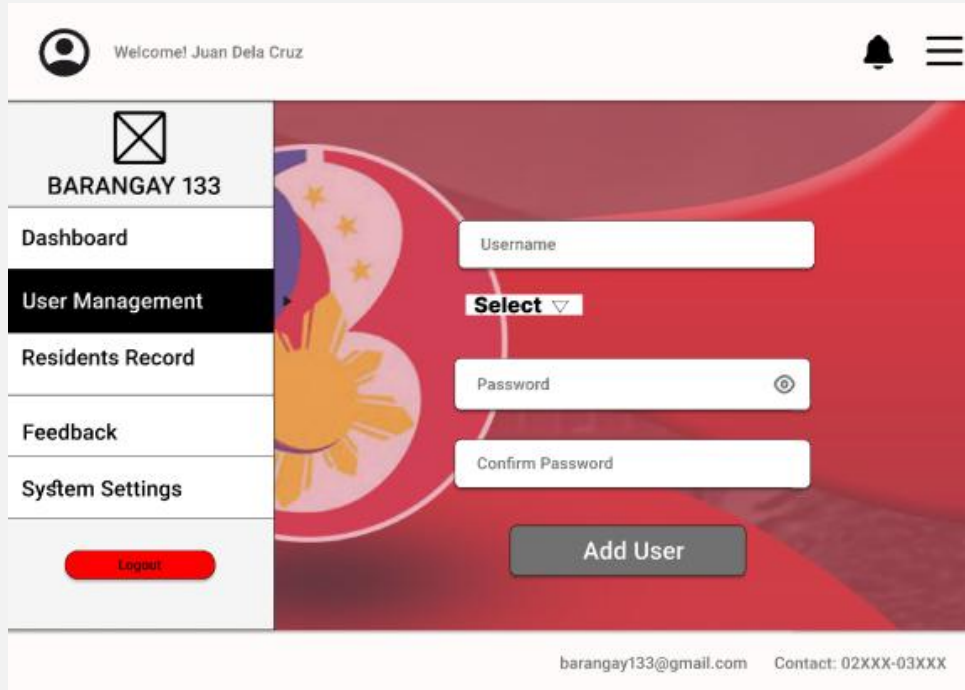


Figure 3.7.6. Add User Form

Figure 3.7.6, the **User Management Interface** enables authorized users to create and manage system accounts within the Barangay 133 system through a single, integrated form. It includes input fields for **Username**, **Role Selection**, **Password**, and **Confirm Password**, ensuring accurate account creation and secure credential setup. A clearly labeled **Add User** button allows administrators to submit new user details efficiently.

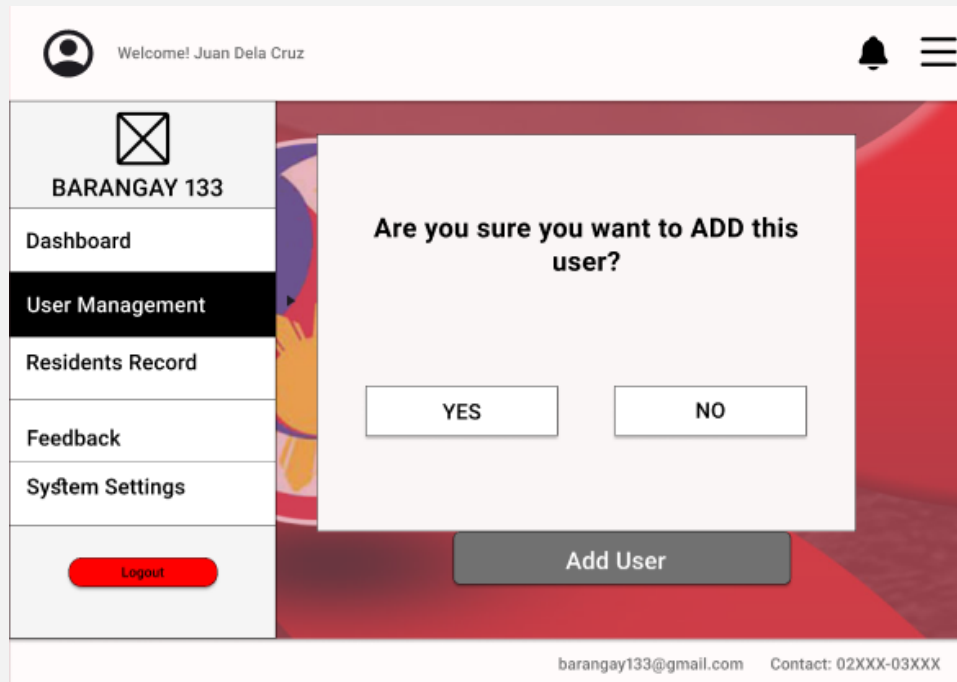


Figure 3.7.7 Add User Confirmation Dialog

Figure 3.7.7 displays a modal Add User Confirmation Dialog, which appears as an overlay on the User Management interface, specifically following the submission of the Add User Form. This dialog serves as a final verification step to ensure the administrator's intent before account creation. The prompt explicitly asks: "Are you sure you want to ADD this user?" The administrator is presented with two mandatory options: "YES" to confirm the details and proceed with creating the new user account, or "NO" to cancel the operation and return to the form for potential modifications. This safeguard adds necessary friction to the account provisioning process, minimizing the risk of incorrectly created accounts.

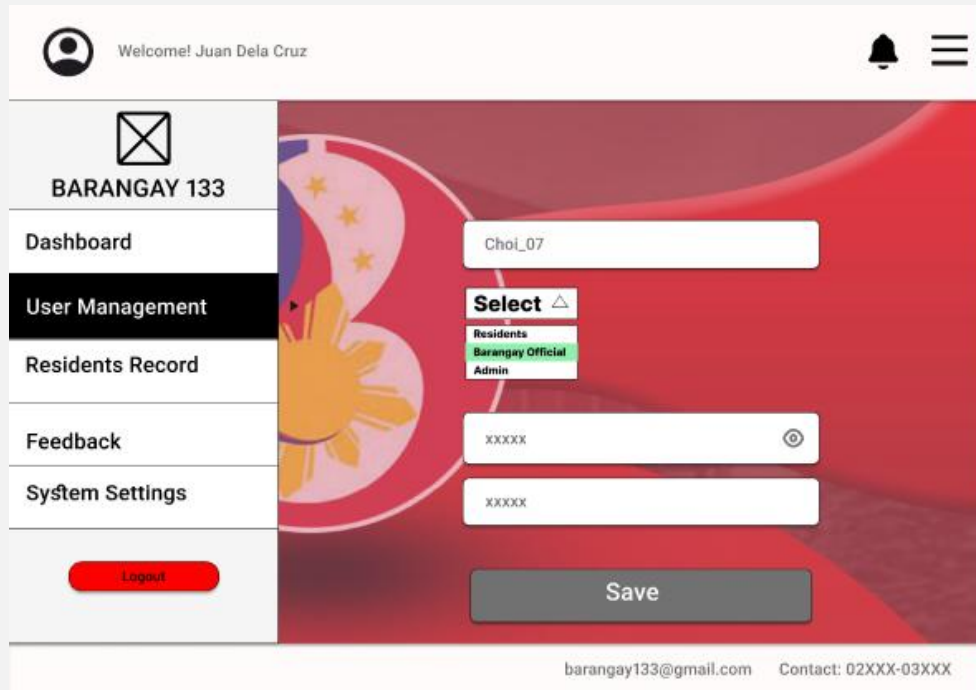


Figure 3.7.8 Edit User Form

Figure 3.7.8 presents the Edit User Form interface prototype, accessible within the User Management module and intended for modifying existing user account details. The layout maintains consistency with the system design, with the Navigation Panel positioned on the left side and clearly highlighting the active User Management section. The Content Area is populated with the current data of a selected user (e.g., "Choi_07"), which is displayed in the Username input field. The form allows administrators to update the Username, change the user's Role through a selection dropdown (with options for Residents, Barangay Official, and Admin), and modify the account credentials using masked Password and Confirm Password fields. A password visibility toggle is present to support secure review during updates. Instead of an "Add" button, the interface concludes with the Save button located at the bottom of the form, which commits all changes made to the existing user record. This form ensures secure and controlled modification of established system accounts while maintaining alignment with the visual layout shown in the interface.

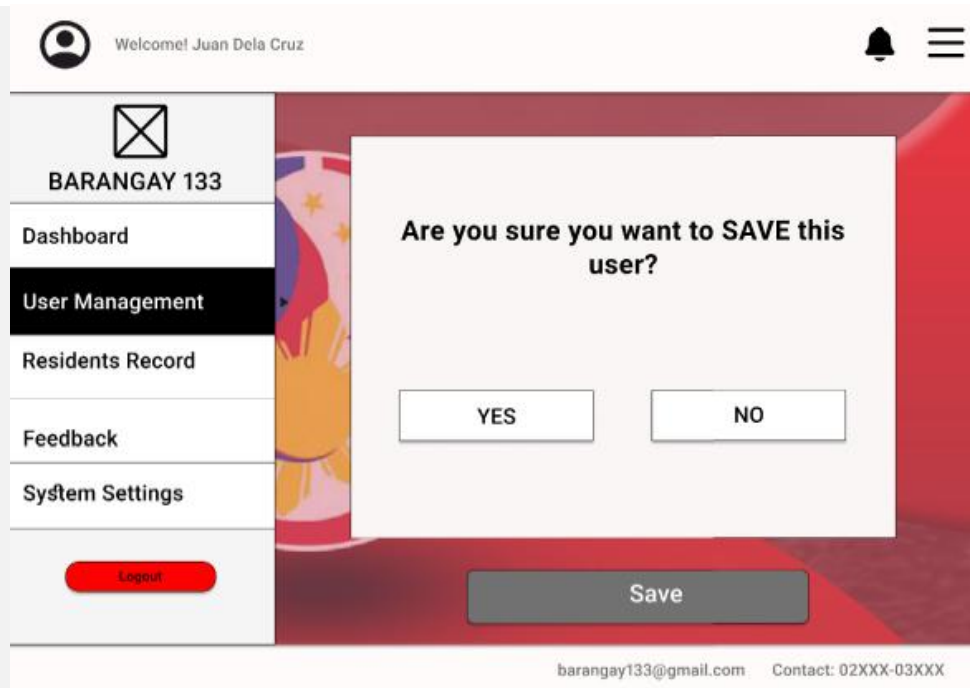


Figure 3.7.9 Edit User Confirmation Dialog

Figure 3.7.9 depicts a modal Edit User Confirmation Dialog, which is activated upon submitting changes via the Edit User Form. This dialog serves as a crucial system safeguard to verify the administrator's intent before updating an existing user record. The prompt explicitly asks: "Are you sure you want to SAVE this user?" The administrator must choose between two options: "YES" to confirm the modifications and apply them to the user's profile, or "NO" to discard the changes and return to the editing form. This final verification step prevents unintended changes to system accounts.

Welcome! Juan Dela Cruz

X

BARANGAY 133

Dashboard

User Management

Residents Record

Feedback

System Settings

Logout

Search users...

Add New Resident

	First Name	Middle Name	Last Name	Gender	Address
1	Sunday	Lozano	Nadonga	Female	1213 dyan lang St.
2	Ji-Hoon	N/A	Jung	Male	1213 dyan lang St.

Next

barangay133@gmail.com

Contact: 02XXX-03XXX

Figure 3.7.10 Residents Records

Figure 3.7.10 displays the Residents Records interface prototype, the module dedicated to managing the demographic and contact information of all registered barangay residents. The standard Navigation Panel is maintained, with the Residents Records link visually highlighted to denote the active view. The Content Area is structured around a central table for data management. Key administrative features include a Search Bar for locating specific individuals, a Filter by Role dropdown (though its utility here may relate to different resident categories), and an Add New User button for registering new residents. The primary feature is the table, which lists resident records with columns providing essential data points: First Name, Middle Name, Last Name, Gender, and Address. This module ensures administrators can efficiently maintain, search, and manage comprehensive resident data.

[illegible]

Figure 3.8.0 Residents Records Continuation

Figure 3.8.0, serving as a continuation of **Figure 3.7.10**, illustrates the detailed contact and status data within the Residents Records management module. The interface maintains the standard layout with the Navigation Panel clearly highlighting the active Residents Records section. The Content Area supplements the demographic data with additional resident attributes, presented within the main table. The columns displayed are Mobile Number, Birthday, Email, and Civil Status. A final Action column provides controls (Edit and Delete icons) for performing administrative modifications or removals of individual resident records. This continuation confirms the module's capability to manage comprehensive and actionable personal resident data.

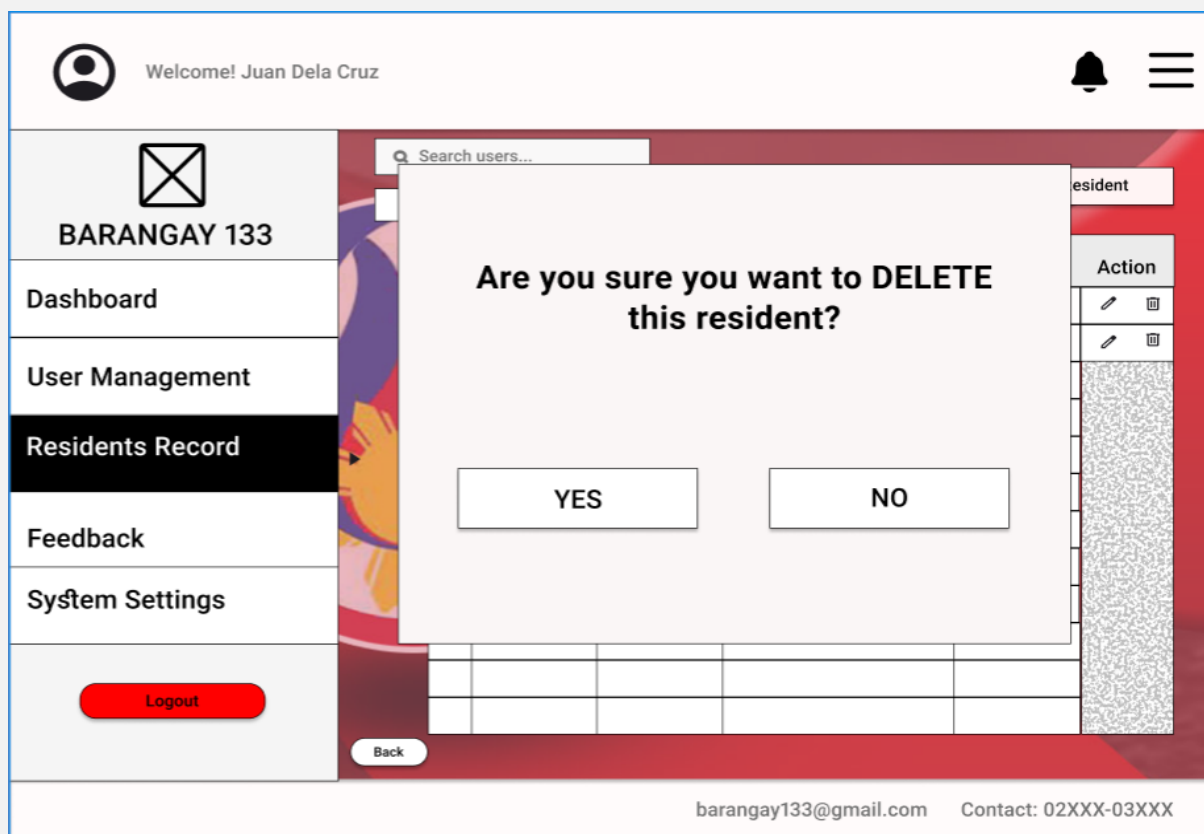


Figure 3.8.1 Deletion of Resident Confirmation

Figure 3.8.1 illustrates the system's critical safety mechanism designed to prevent accidental data loss during administrative maintenance. The interface displays a modal dialog box that is triggered immediately after an administrator attempts to remove an entry from the Residents Record database. This confirmation prompt serves as a final validation step, explicitly querying the user, "Are you sure you want to DELETE this resident?" while offering binary decision controls—"YES" to execute the permanent removal of the record, or "NO" to abort the operation. This design pattern enforces data integrity by ensuring that irreversible actions, such as the deletion of resident profiles, are performed intentionally and with explicit verification.

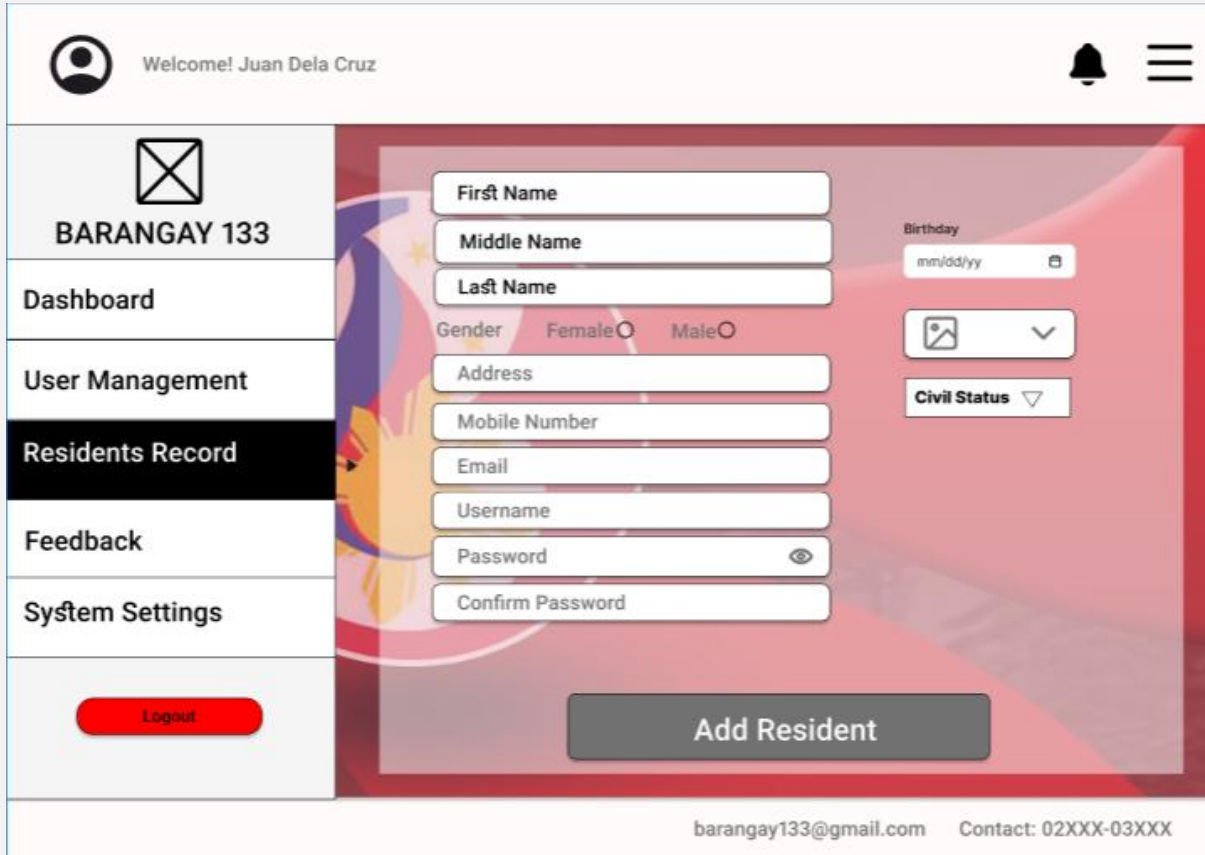


Figure 3.8.2 Add Resident Form

Figure 3.8.2 presents the Add Resident Form; the primary data entry interface designed for the registration of new community members into the system. Accessible via the "Residents Record" administrative module, this form serves as the front-end mechanism for populating the resident database, requiring administrators to input comprehensive demographic details including the individual's full name, gender, birth date, civil status, and physical address. Beyond personal identification, the interface simultaneously captures essential contact information (mobile number, email) and establishes system access credentials through username and password fields, the latter of which includes a confirmation step to ensure input accuracy. The process concludes with the "Add Resident" command button, which triggers the validation and subsequent transmission of the encoded data to the central database server for permanent storage and account activation.

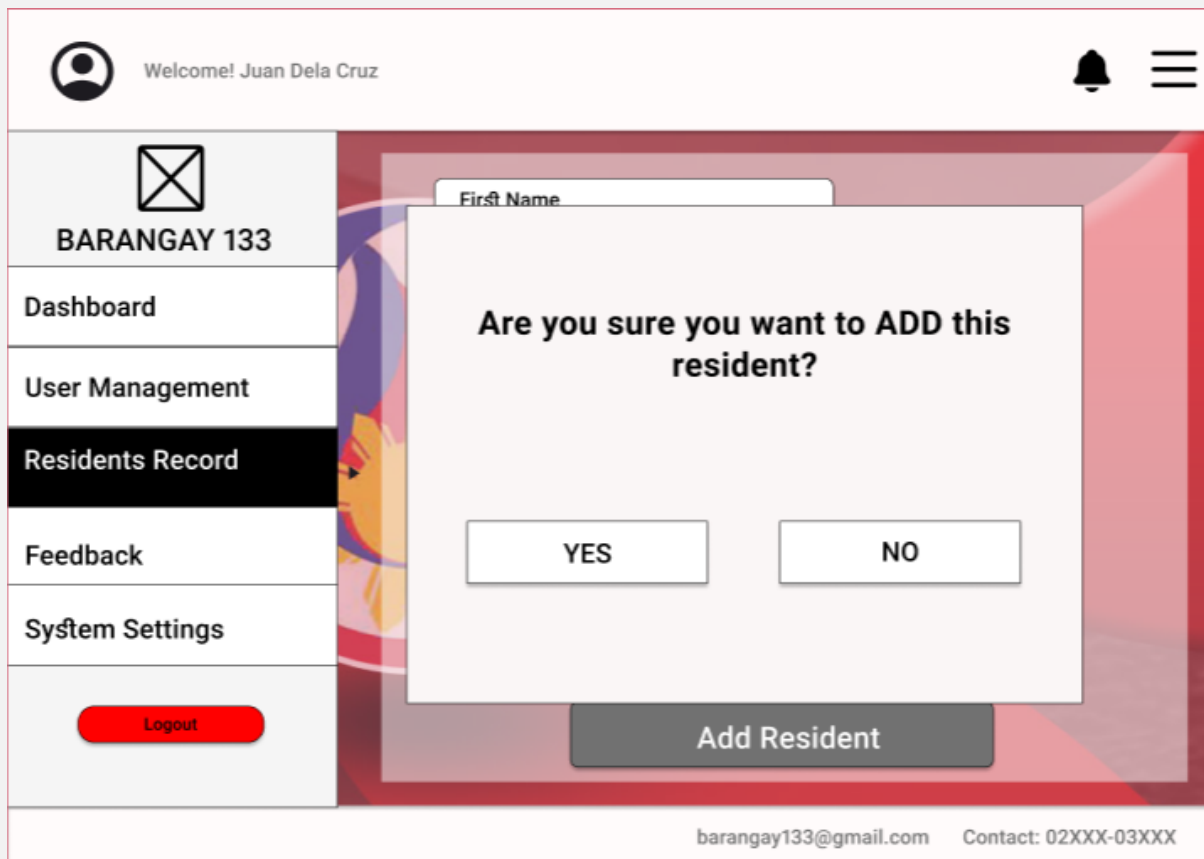
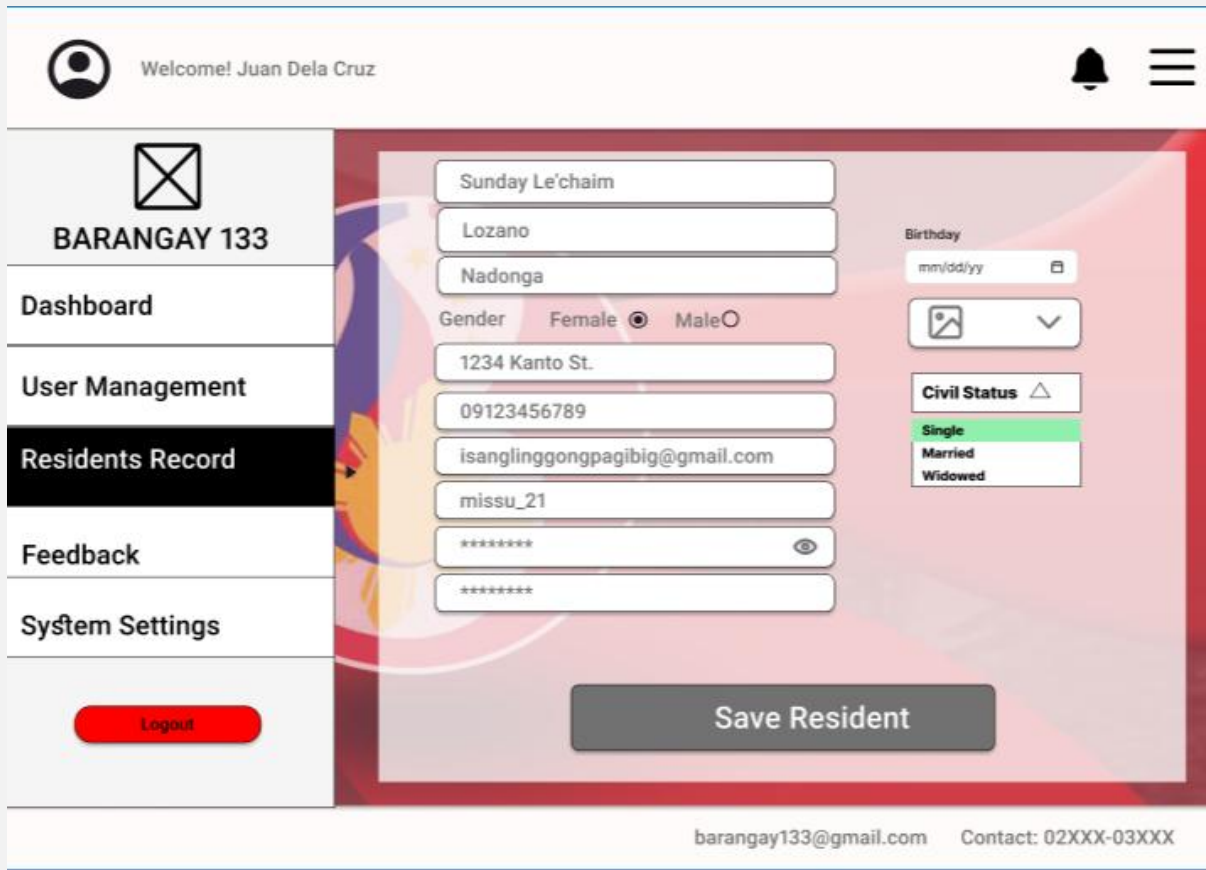


Figure 3.8.3 Add Resident Confirmation

Figure 3.8.3 depicts the system's data validation mechanism, specifically the Add Resident Confirmation dialog. This modal interface is triggered immediately upon the submission of the resident registration form, acting as a final procedural checkpoint before data commitment. The interface presents a clear query, "Are you sure you want to ADD this resident?" accompanied by binary "YES" and "NO" options. This design pattern is implemented to mitigate human error, ensuring that the administrator explicitly verifies the intent to register the new profile into the database, thereby maintaining the integrity and accuracy of the resident records.



Welcome! Juan Dela Cruz

BARANGAY 133

Dashboard

User Management

Residents Record

Feedback

System Settings

Logout

Sunday Le'chaim

Lozano

Nadonga

Gender ☒ Female ☐ Male

1234 Kanto St.

09123456789

isanglinggongpagibig@gmail.com

missu_21

Birthdate mm/dd/yy

Civil Status ☒ Single ☐ Married ☐ Widowed

Save Resident

barangay133@gmail.com Contact: 02XXX-03XXX

Figure 3.8.4 Edit Resident Form

Figure 3.8.4 illustrates the Edit Resident Form, an administrative interface designed to ensure the continuous accuracy and relevance of the demographic database. This module is invoked when an administrator selects a specific resident record for modification, prompting the system to retrieve and auto-populate the input fields with the existing stored data—such as the resident's full name, physical address, contact details, and civil status. This design facilitates efficient data maintenance, allowing administrators to seamlessly rectify typographical errors or update personal information (e.g., change of address or marital status) without re-entering unchanged data. The modification process is concluded by the "Save Resident" command, which triggers an update transaction to overwrite the previous record in the central database with the newly validated information.



Figure 3.8.5 Save Resident Confirmation

Figure 3.8.5 illustrates the Save Resident Confirmation interface; a critical validation checkpoint integrated into the resident modification workflow. Following the administrator request to update a resident profile, this modal dialog is actuated to solicit explicit consent before committing changes to the database. The interface presents the query, "Are you sure you want to SAVE this resident?" coupled with binary "YES" and "NO" command options. This design pattern ensures data integrity by preventing accidental overwrites or unintended modifications, requiring the administrator to definitively affirm the accuracy of the edited information before the system executes the permanent update transaction.

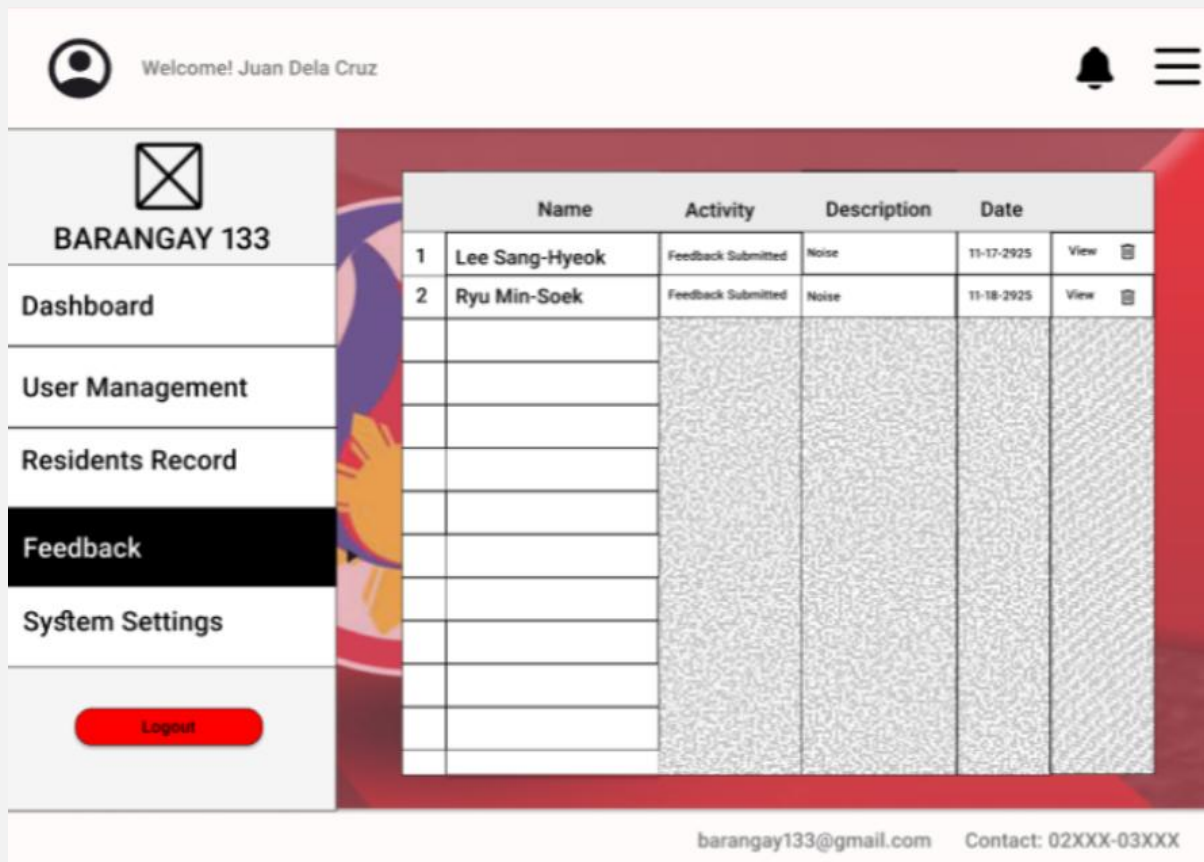


Figure 3.8.6 Super Admin Feedback Dashboard

Figure 3.8.6 displays the Super Admin Feedback Dashboard; a centralized monitoring interface designed to aggregate and manage community-driven reports. Located within the "Feedback" module, this view presents data in a structured tabular format, allowing administrators to efficiently scan through resident submissions. The table categorizes each entry by the resident's Name (e.g., Lee Sang-Hyeok, Ryu Min-Soek), the specific Activity type (e.g., "Feedback Submitted"), a brief Description of the concern (such as "Noise" complaints), and the Date of submission. To facilitate administrative action, the interface includes specific controls for each row, enabling the administrator to either "View" the full details of a specific complaint or "Delete" resolved or irrelevant entries, thereby ensuring the feedback loop remains organized and actionable.

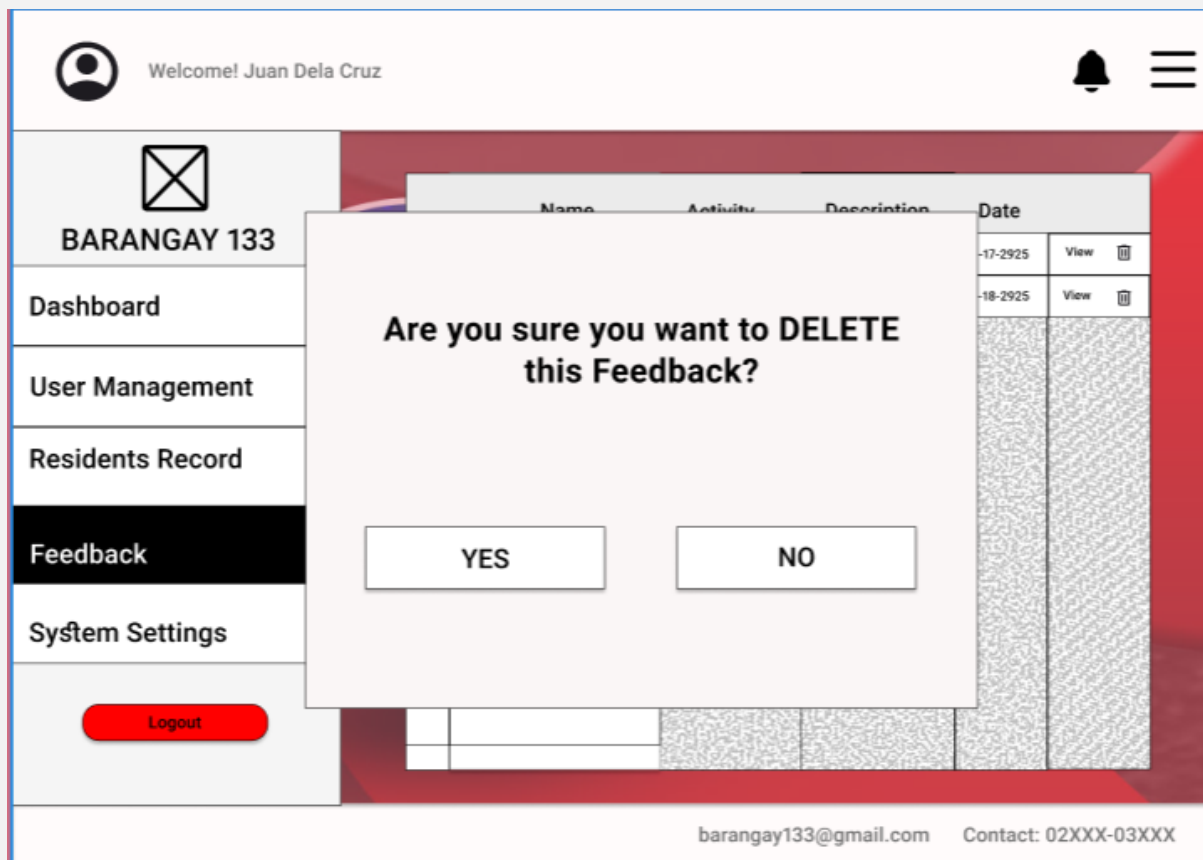


Figure 3.8.7 Deletion of Feedback Confirmation

Figure 3.8.7 depicts the Deletion of Feedback Confirmation interface, a safety protocol implemented within the system's feedback management module. Triggered when an administrator attempts to remove a specific entry from the feedback log, this modal dialog acts as a mandatory verification step to prevent accidental loss of community data. The interface displays a prominent warning query, "Are you sure you want to DELETE this Feedback?" and provides binary "YES" and "NO" options. This design ensures that the permanent removal of resident complaints or suggestions is a deliberate and confirmed action, thereby maintaining the accuracy and integrity of the administrative records.

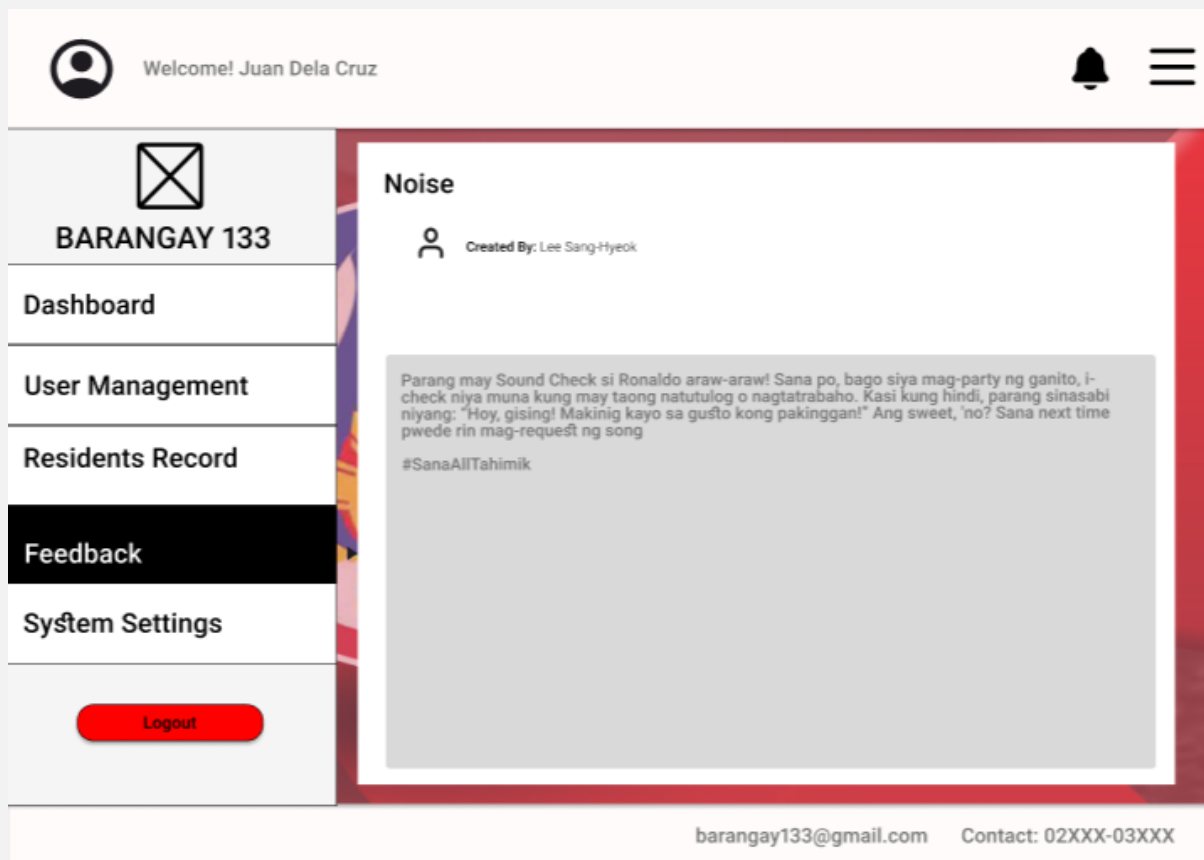


Figure 3.8.8 View Residents Feedback

Figure 3.8.8 presents the View Residents Feedback interface, a specialized detail view designed to provide the Super Admin with the comprehensive context of a specific resident report. Accessible via the main feedback dashboard, this screen displays the full qualitative data of a submission, prominently featuring the Subject (e.g., "Noise") and the Author's Identity (e.g., "Created By: Lee Sang-Hyeok"). The core function of this interface is to render the complete body of the resident's message, enabling the Super Admin to review detailed narratives regarding community issues, such as noise complaints or neighborhood disturbances, thereby facilitating a high-level assessment of community concerns before directing them to the appropriate officials or taking system-level action.

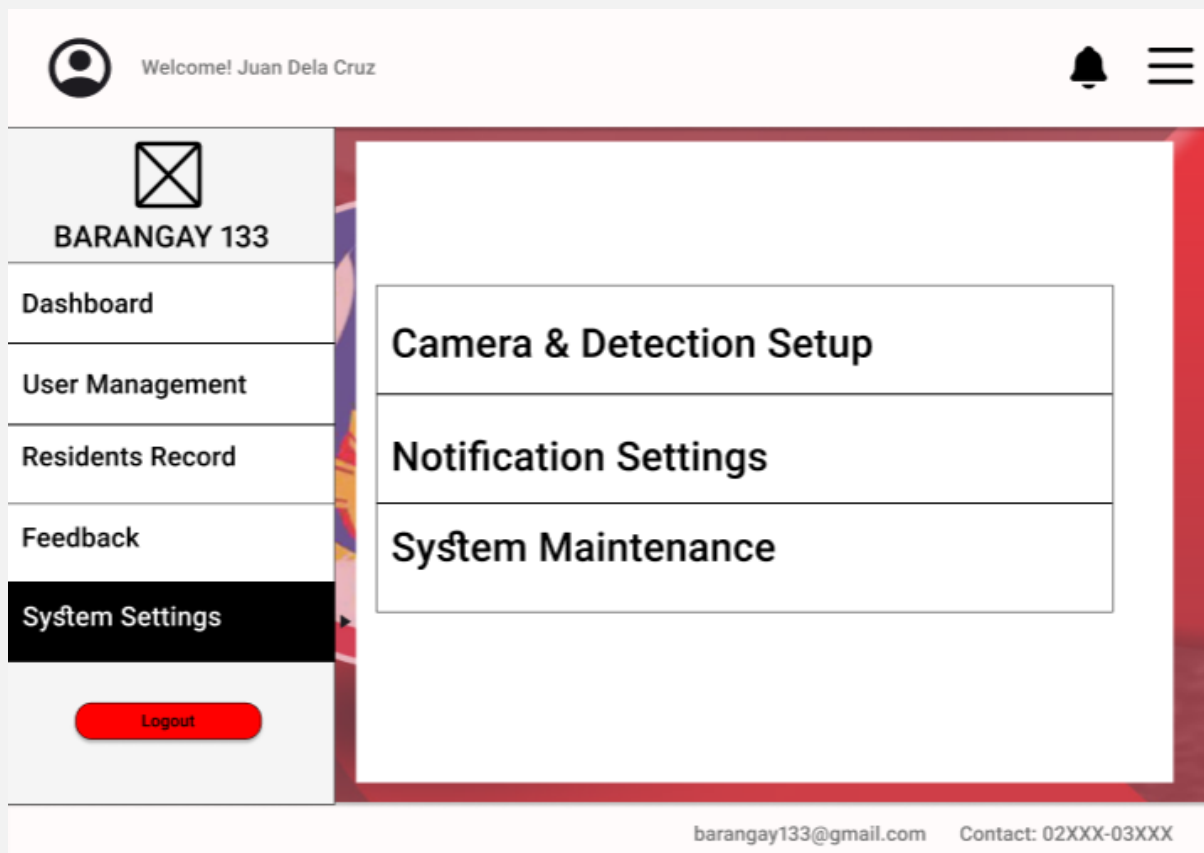


Figure 3.8.9 System Settings

Figure 3.8.9 depicts the System Settings dashboard; a privileged interface designed for the comprehensive management of the application of core technical parameters. Accessible only to high-level administrators (Super Admins); this module categorizes configuration options into three distinct functional domains to streamline system administration. The "Camera & Detection Setup" section facilitates the calibration and management of the hardware integration and image processing algorithms used for garbage detection; "Notification Settings" governs the protocols and triggers for disseminating alerts to residents; and "System Maintenance" provides essential tools for routine database optimization and software upkeep. This centralized configuration hub ensures that critical operational variables can be adjusted dynamically to maintain system stability without requiring direct code intervention.

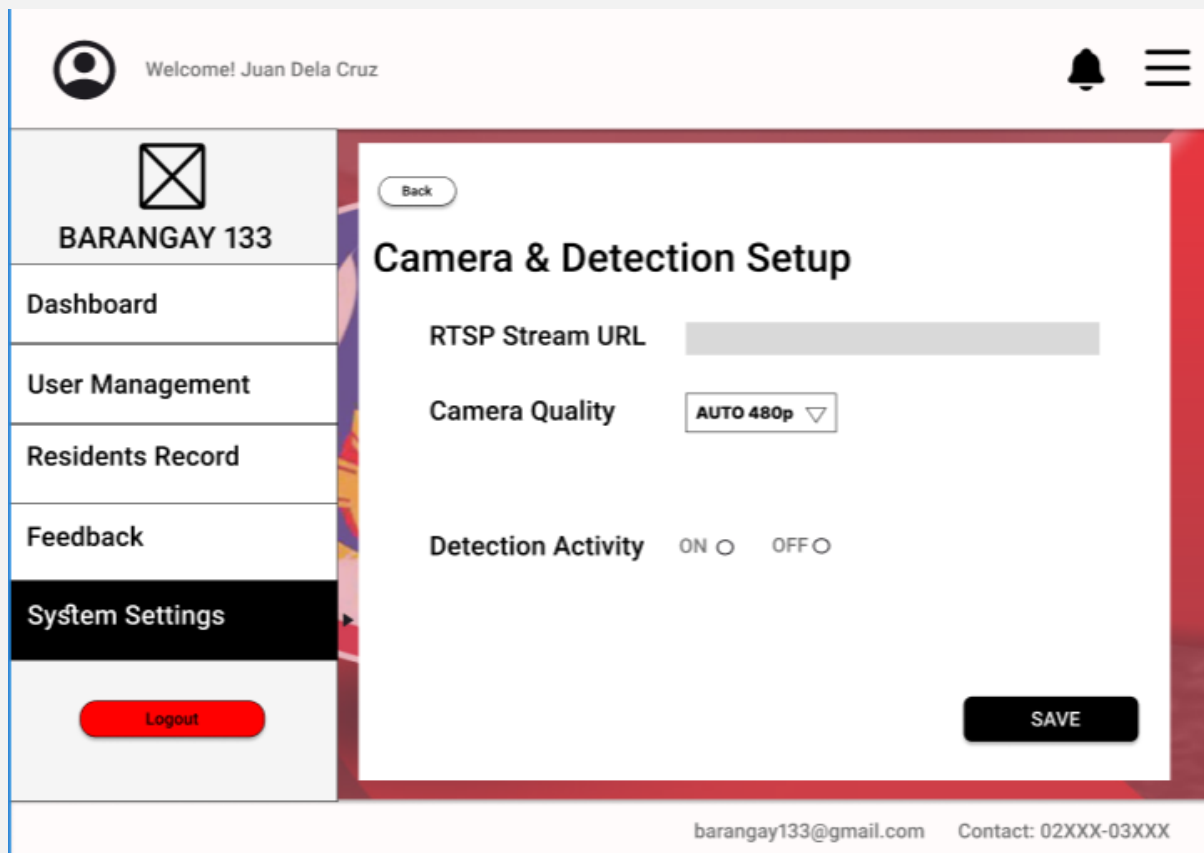


Figure 3.8.10 Camera & Detection Setup

Figure 3.8.10 presents the Camera & Detection Setup interface, a specialized configuration panel within the System Settings module designed to bridge the physical surveillance hardware with the system's software logic. This interface allows the Super Admin to define critical technical parameters for the image processing unit, primarily through the RTSP Stream URL field, which designates the specific network address of the connected IP camera for real-time video feed ingestion. Additionally, the interface provides controls for resource management, including a Camera Quality dropdown to adjust video resolution (e.g., 480p) for bandwidth optimization, and a Detection Activity toggle (ON/OFF) that allows the administrator to manually enable or disable the automated garbage detection algorithm as operational requirements dictate.

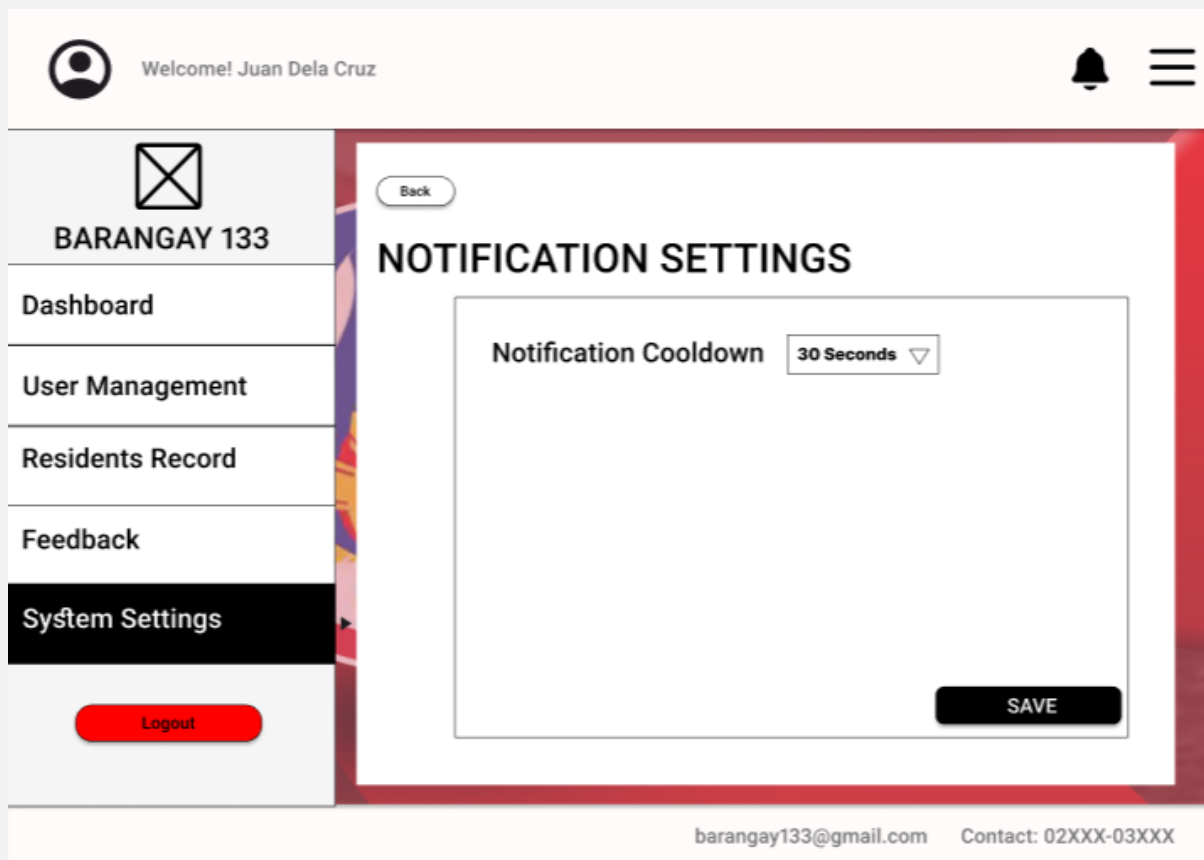


Figure 3.9.0 Notification Settings

Figure 3.9.0 displays the Notification Settings interface, a sub-module of the system configuration designed to regulate the frequency of automated alerts. The primary control mechanism shown is the Notification Cooldown parameter, which allows the Super Admin to enforce a mandatory time interval (depicted here as "30 Seconds") between successive notifications. This feature is critical for user experience optimization, preventing "alert fatigue" by ensuring that residents are not overwhelmed with repetitive messages during prolonged or continuous detection events. The interface is streamlined to focus solely on this operational constraint, providing a straightforward "SAVE" command to commit the timing adjustments to the backend logic.

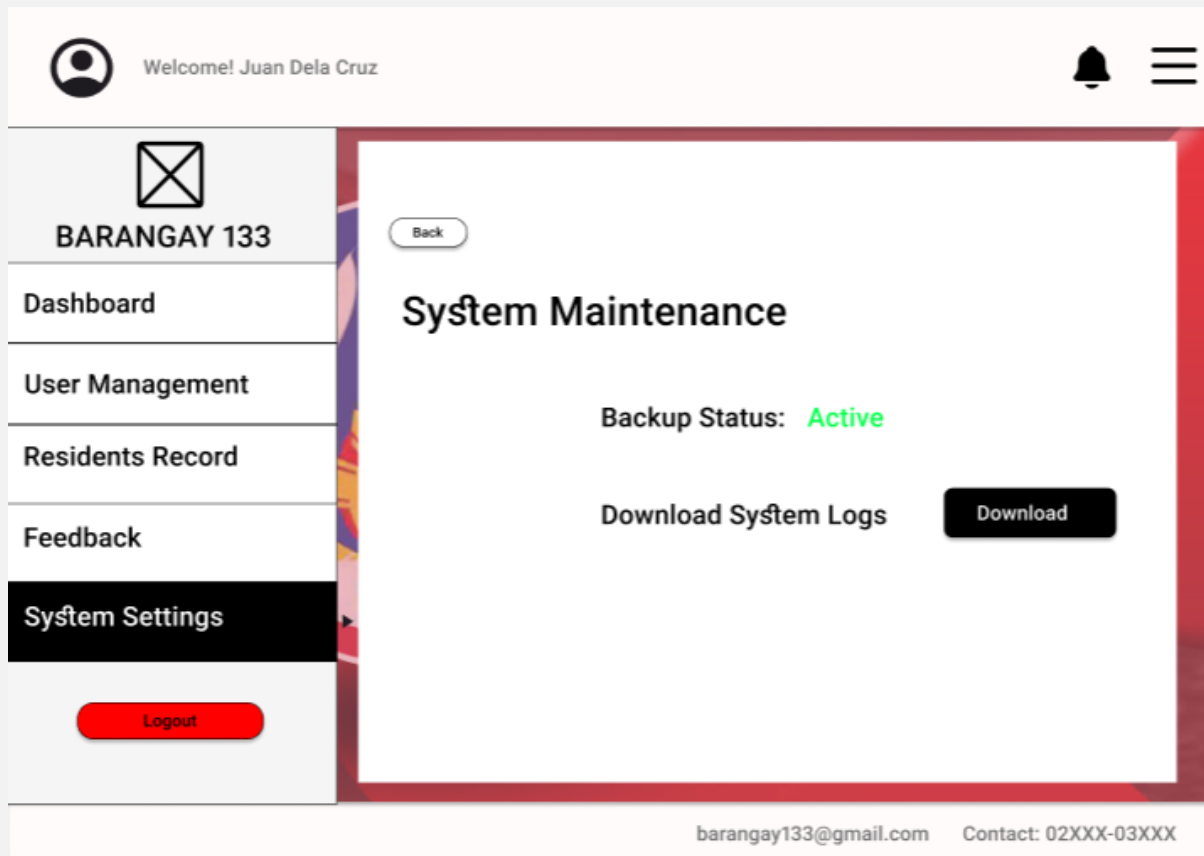


Figure 3.9.1 System Maintenance

Figure 3.9.1 illustrates the System Maintenance interface; a critical administrative utility designed to ensure data preservation and facilitate system auditing. This screen provides the Super Admin with a real-time status indicator of the automated data protection protocols, prominently displaying the Backup Status as "Active" to confirm that redundancy measures are operational. Furthermore, the interface includes a dedicated Download System Logs facility controlled by a "Download" command button. This feature enables the extraction of comprehensive activity logs from the server, allowing administrators to perform offline troubleshooting, security analysis, or compliance reporting by retrieving a chronological record of system events.

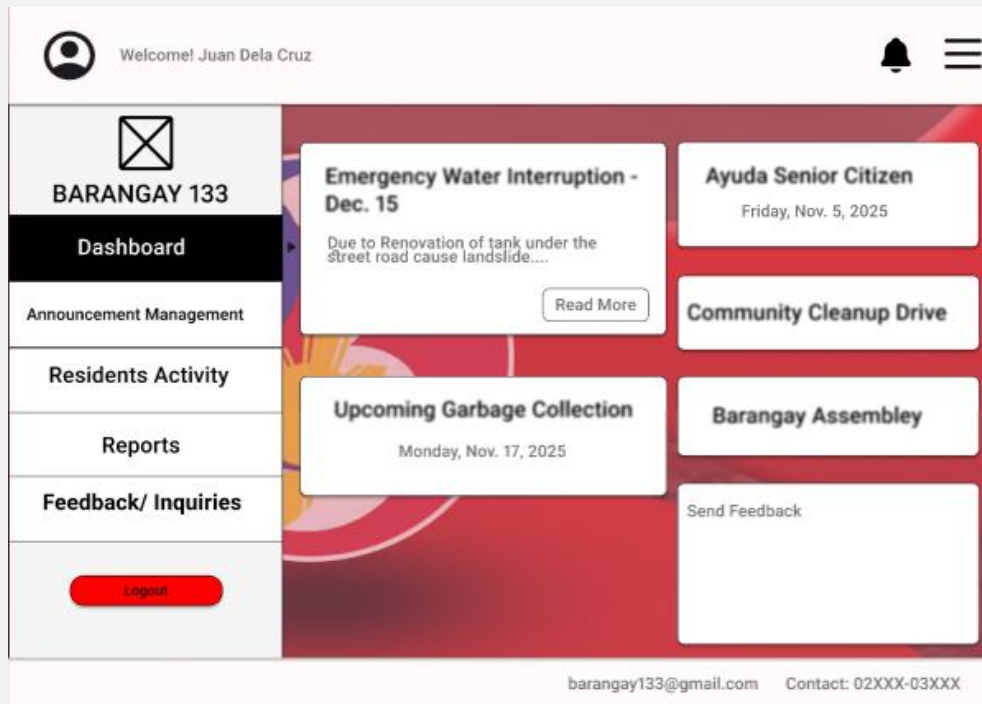


Figure 3.9.2 User Dashboard

Figure 3.9.2 shows the **User Dashboard**, which serves as the main landing page after successful login. It displays the **BARANGAY 133** logo and a personalized welcome message at the top, along with a left-side navigation panel for accessing system modules. The main content area presents important barangay announcements, scheduled services, and community events in card-based sections, as well as a **Send Feedback** field for user interaction.

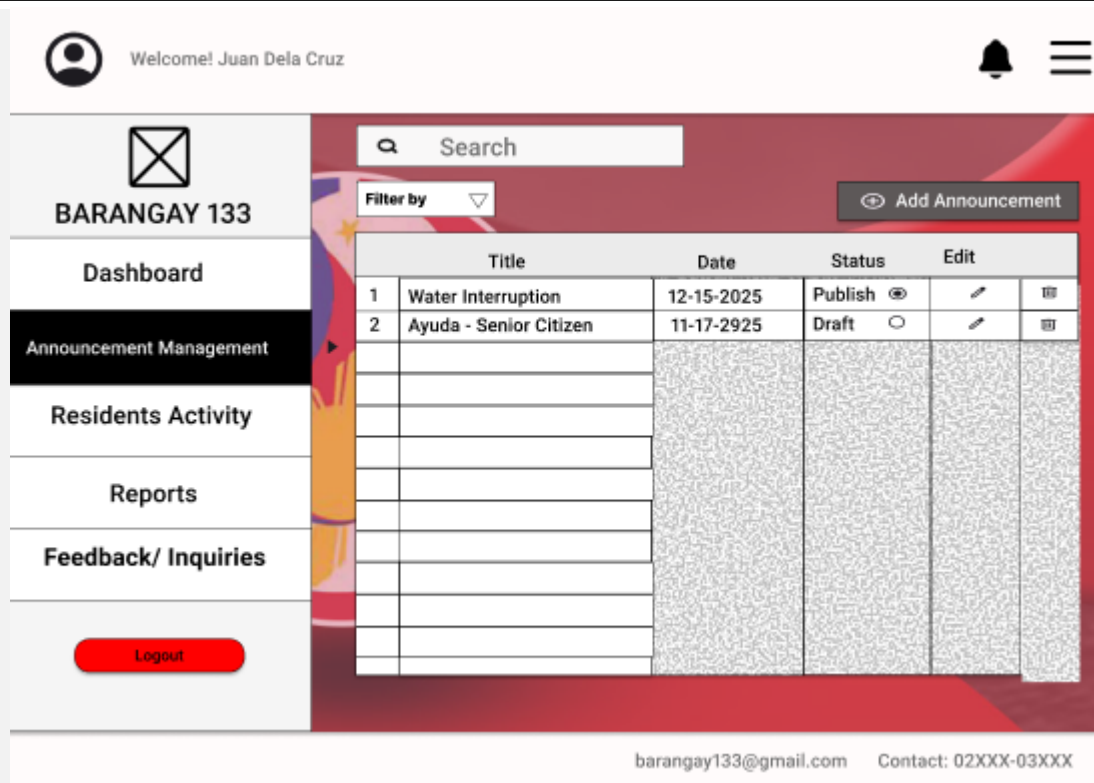


Figure 3.9.3. Announcements Management Page

Figure 3.9.3 displays the system's dedicated Announcements Management Page, the administrative hub for controlling community updates. It maintains a clean and consistent layout with the sidebar navigation clearly highlighting the active section. The key function of this page is efficient content administration, supported by a search bar, a "Filter by Date" option, and a prominent "Add Announcement" button for creating new posts. Existing records are displayed clearly in a tabular format showing titles (e.g., "Water Interruption"), dates, and current status (e.g., "Publish" or "Draft"), along with direct action buttons to edit or delete specific items.

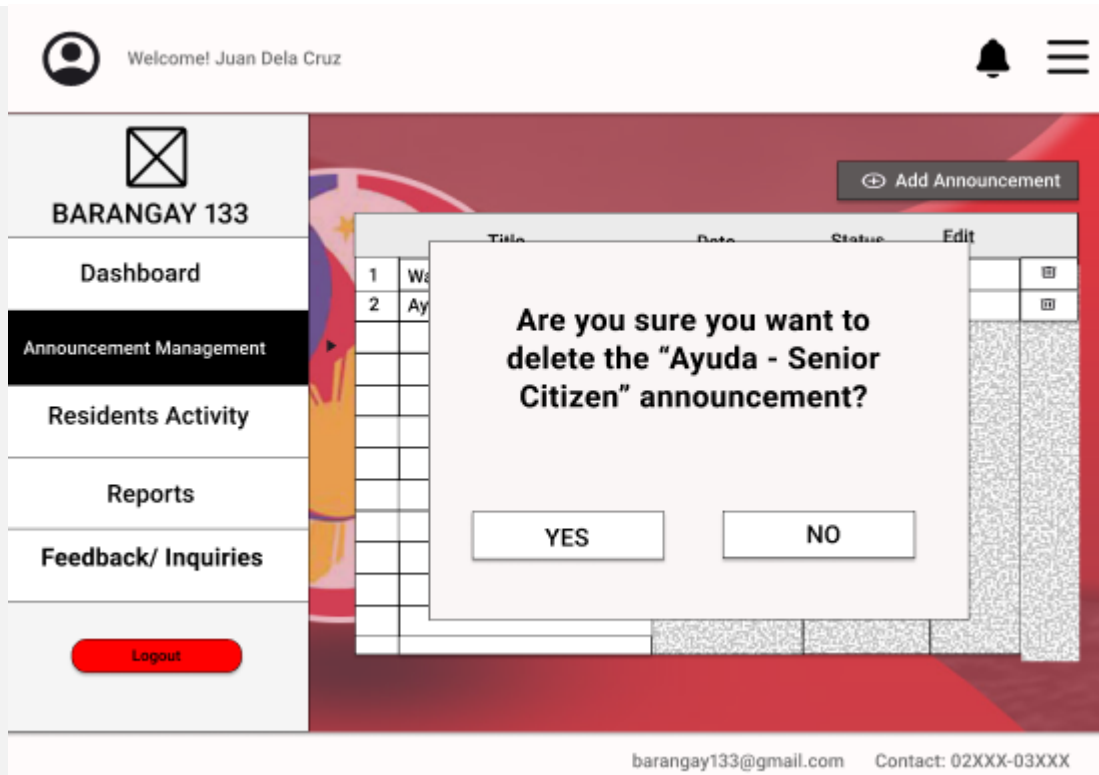


Figure 3.9.4. Delete Confirmation Modal

Figure 3.9.4 displays the system's Delete Confirmation Modal, a critical safety mechanism triggered within the Announcment Management page. It ensures data integrity by preventing the accidental removal of official records. The key function of this interface is to validate the user's intent, appearing as a prominent overlay that temporarily blocks interaction with the main table. The modal clearly presents a specific warning query, "Are you sure you want to delete the 'Ayuda - Senior Citizen' announcement?" and provides distinct "YES" and "NO" buttons to either execute the permanent deletion or cancel the action.

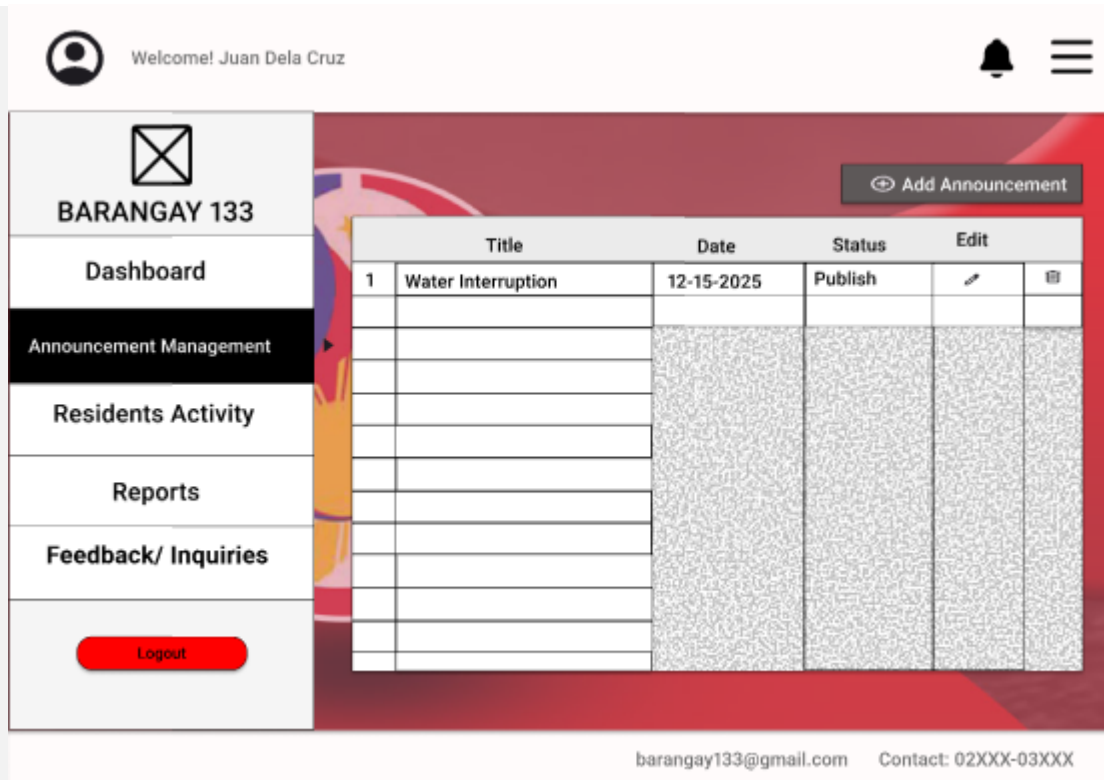
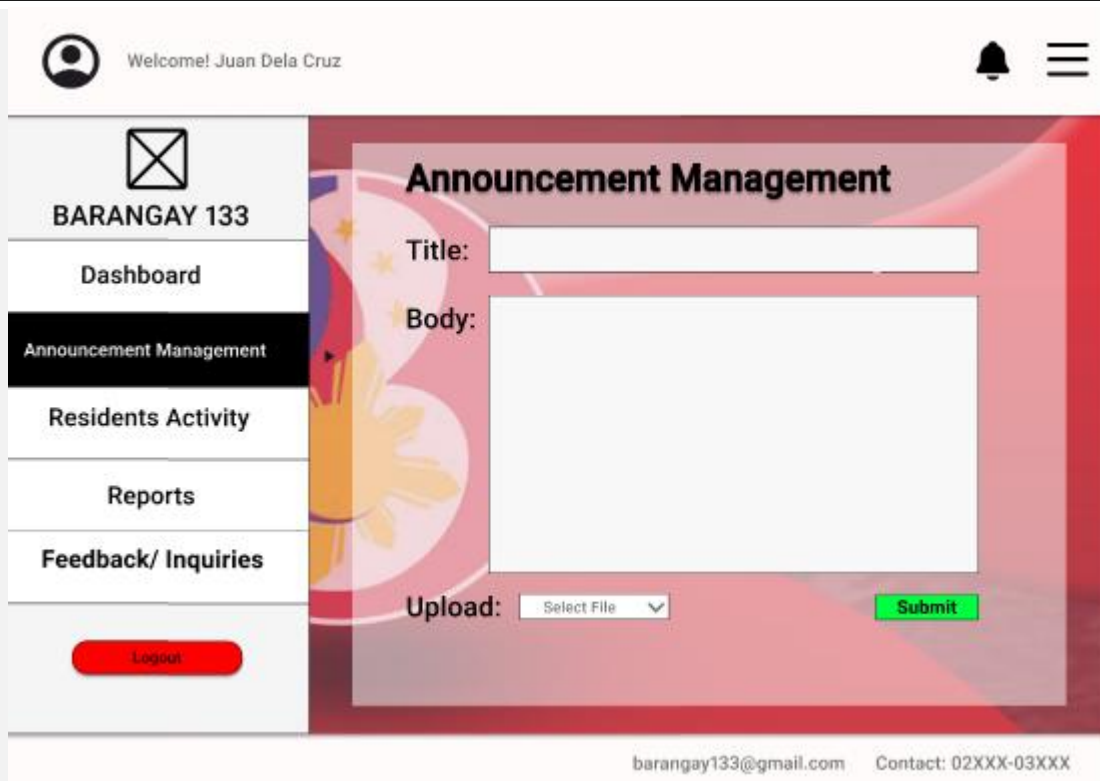


Figure 3.9.5 Announcement Management Interface

Figure 3.9.5, the Announcement Management Interface is designed to allow authorized users to create, view, and manage barangay announcements within the system. The interface follows a consistent layout, featuring a left-side navigation panel that highlights the active Announcement Management module for easy navigation. The main content area displays announcements in a tabular format, providing structured information such as the Title, Date, and Status of each announcement. Action icons are included in the Edit column, enabling administrators to modify or update existing announcements efficiently. An Add Announcement button is positioned at the upper section of the interface, allowing users to create new announcements as needed.



The screenshot displays a web application interface for 'Announcment Management'. At the top, a user profile icon is followed by the text 'Welcome! Juan Dela Cruz'. To the right are a bell icon and a hamburger menu icon. A left-side navigation panel contains a header 'BARANGAY 133' with a square icon, and a list of menu items: 'Dashboard', 'Announcment Management' (highlighted in black), 'Residents Activity', 'Reports', and 'Feedback/ Inquiries'. A red 'Logout' button is at the bottom of the panel. The main content area has a red background with a decorative floral pattern. It features a form titled 'Announcment Management' with a 'Title:' label and a text input field, a 'Body:' label and a larger text area, and an 'Upload:' section with a 'Select File' button and a green 'Submit' button. The footer shows the email 'barangay133@gmail.com' and the contact number '02XXX-03XXX'.

Figure 3.9.6 Announcment Management Interface

Figure 3.9.6, this figure presents the Announcment Management – Add Announcment Form, which allows authorized users to create and publish new barangay announcements. The interface maintains consistency with the system layout, featuring a left-side navigation panel with the Announcment Management section highlighted. The main content area contains input fields for the Title and Body of the announcement, enabling administrators to enter detailed and relevant information. An Upload option is provided to attach supporting files or images to the announcement. A Submit button is positioned at the lower right of the form to finalize and publish the announcement.

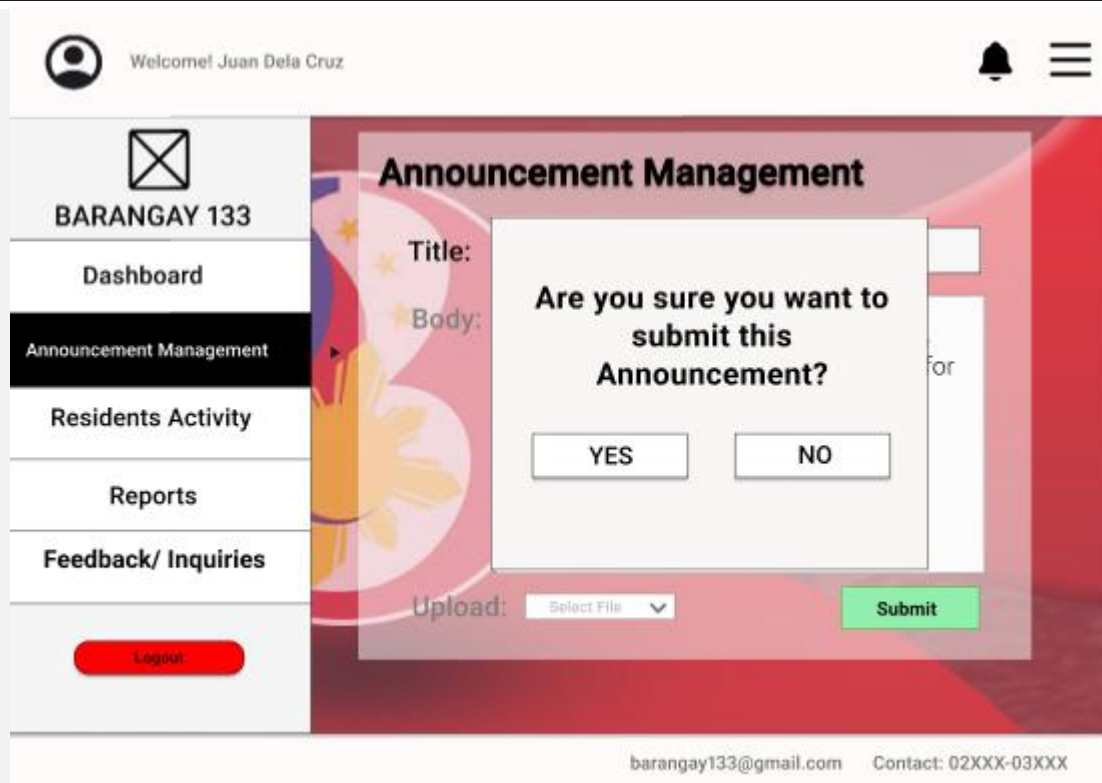
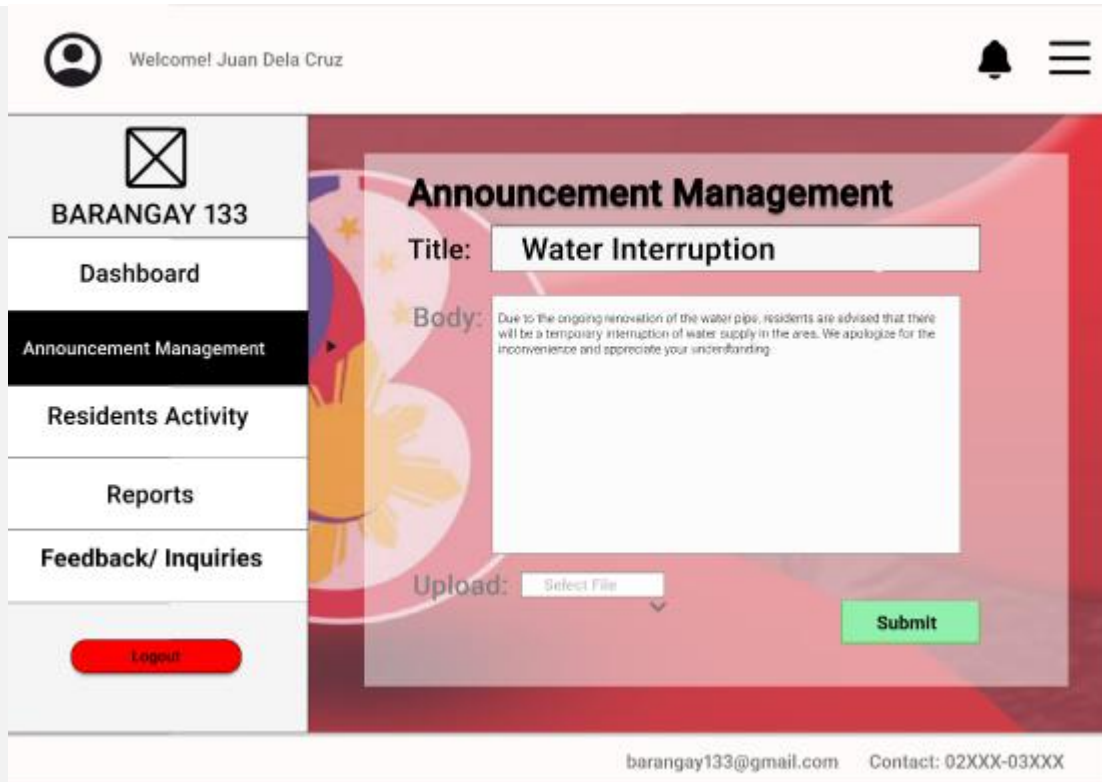


Figure 3.9.7 Announcement Submission Confirmation Dialog

Figure 3.9.7, this figure presents the Announcement Submission Confirmation Dialog, which appears when a user attempts to submit a new announcement within the Announcement Management module. The dialog is displayed as a pop-up overlay on top of the announcement form to prompt user confirmation before final submission. The message, *"Are you sure you want to submit this Announcement?"*, ensures that users intentionally proceed with publishing the announcement. Two action buttons, YES and NO, are provided to allow users to either confirm the submission or cancel the action and return to editing the announcement details. This confirmation step helps prevent accidental submissions and enhances the reliability of the announcement management process.



BARANGAY 133

Dashboard

Announcement Management

Residents Activity

Reports

Feedback/Inquiries

Logout

Announcement Management

Title: **Water Interruption**

Body: Due to the ongoing renovation of the water pipe, residents are advised that there will be a temporary interruption of water supply in the area. We apologize for the inconvenience and appreciate your understanding.

Upload:

barangay133@gmail.com Contact: 02XXX-03XXX

Figure 3.9.8 Add Announcement Form

Figure 3.9.8 displays the system's Add Announcement Form; the interface used for composing and publishing new community updates. It maintains the clean and consistent admin dashboard layout with the sidebar navigation. The key function of this page is content creation, shown here with the fields populated and ready for submission. The layout features a "Title" field (filled with "Water Interruption") and a "Body" section containing specific details about infrastructure maintenance, accompanied by an "Upload" option and a prominent "Save" button to finalize and broadcast the post to residents.

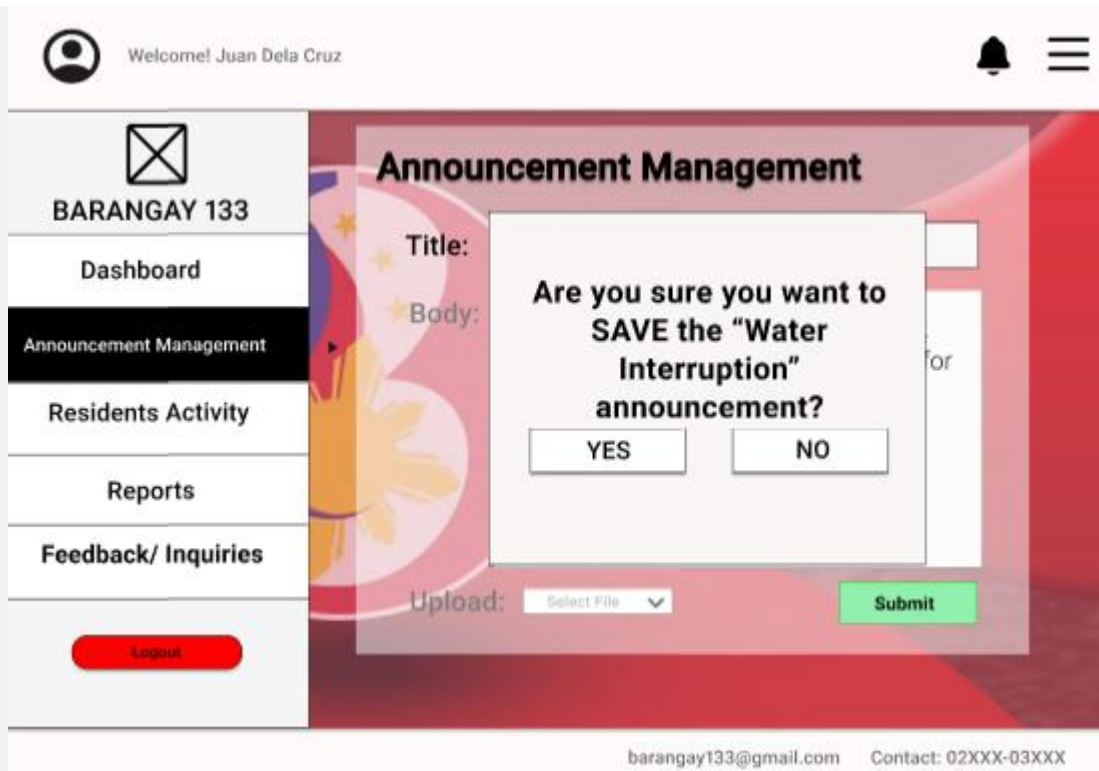


Figure 3.9.9. Save Confirmation Modal

Figure 3.9.9 displays the system's Save Confirmation Modal, a final verification step triggered within the Announcement Management page. It maintains the admin workflow by overlaying the active creation form, ensuring the user does not lose context. The key function of this prompt is to prevent the accidental publication of unverified information. The modal presents a clear query, "Are you sure you want to SAVE the 'Water Interruption' announcement?" and requires the official to explicitly select "YES" to broadcast the update or "NO" to return to the editor.

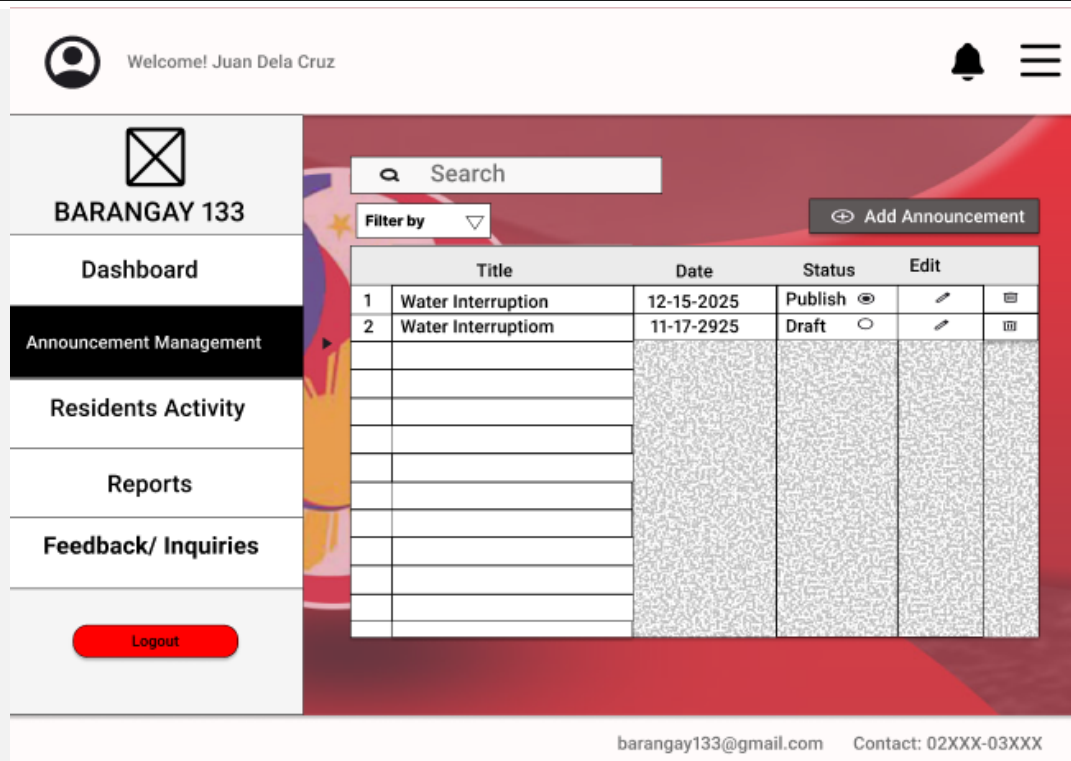


Figure 3.10.0 Announcment Management Page

Figure 3.10.0 displays the system's dedicated Announcment Management Page, the administrative hub for controlling community updates. It maintains a clean and consistent layout with the sidebar navigation, clearly highlighting the active section. The key function of this page is efficient content administration, supported by a search bar, a "Filter by Date" option, and a prominent "Add Announcment" button for creating new posts. Existing records are displayed clearly in a tabular format showing titles (e.g., "Water Interruption"), dates, and current status (e.g., "Publish" or "Draft"), along with direct action buttons to edit or delete specific items.



Welcome! Juan Dela Cruz






BARANGAY 133

Dashboard

Announcement Management

Residents Activity

Reports

Feedback/ Inquiries

Logout

Residents Activity History

Filter by 

	Name	Activity	Description	Date	Status
1	Lee Sang-Hyeok	Feedback Submitted	Noise	11-17-2925	Sent
2	Mun Hyeon-Jun	Login Attempt Failed	Incorrect password or username	11-17-2925	Failed

barangay133@gmail.com Contact: 02XXX-03XXX

Figure 4.0 Residents Activity History Page

Figure 4.0 displays the system's dedicated Residents Activity History Page, the central hub for monitoring user interactions and system security. It maintains a clean and consistent admin dashboard layout with the sidebar navigation clearly highlighting the active section. The key function of this page is to provide a transparent audit trail of resident actions, supported by a search bar and a "Filter by Date" feature for efficient log retrieval. Historical data is displayed clearly in a tabular format, detailing specific events with columns for "Name" (e.g., Lee Sang-Hyeok), "Activity" (e.g., Feedback Submitted, Login Attempt Failed), "Description", "Date", and "Status", allowing officials to track engagement and troubleshoot access issues.

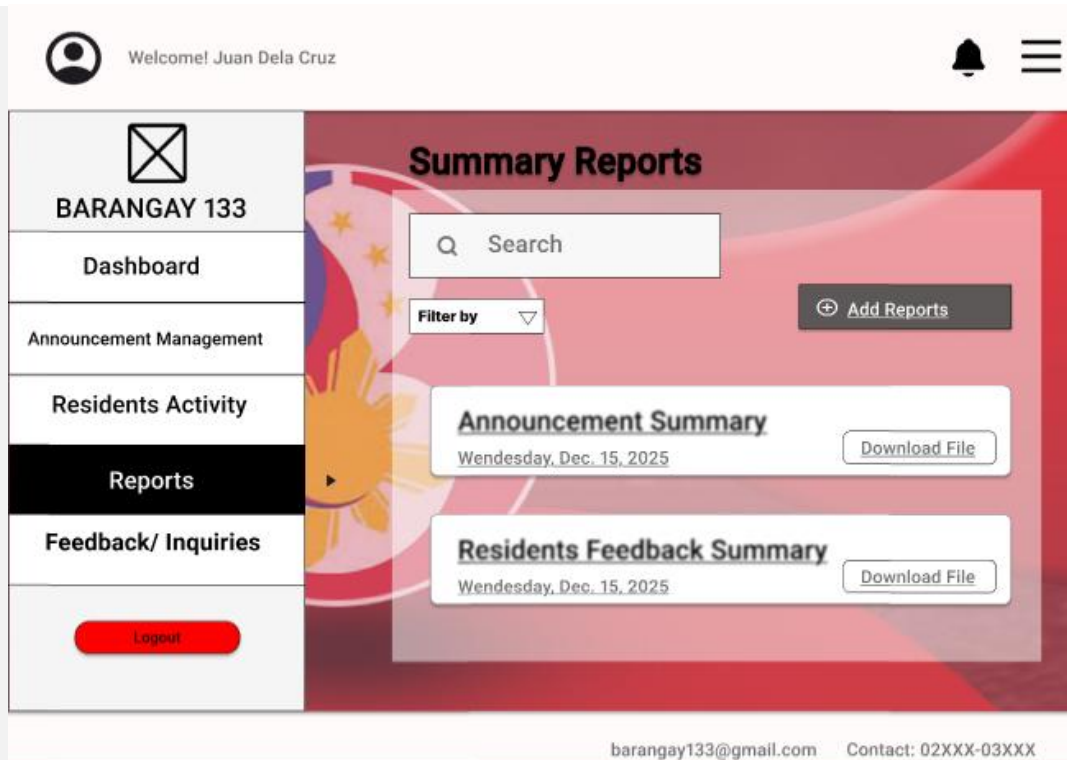
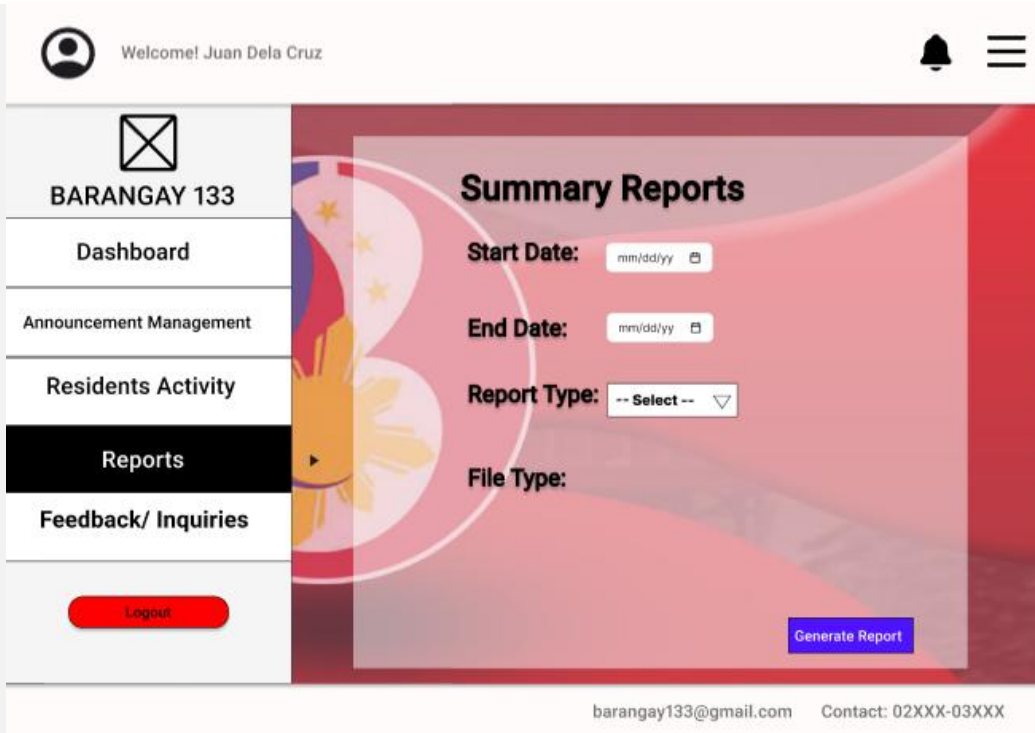


Figure 4.1.0 Summary Reports Page

Figure 4.1.0 displays the system's dedicated Summary Reports Page, the administrative hub for generating and accessing data summaries. It maintains a clean and consistent admin dashboard layout with the sidebar navigation clearly highlighting the active section. The key function of this page is to allow officials to monitor system usage and resident engagement through documented logs, supported by a search bar, a "Filter by Date" option, and an "Add Reports" button. Generated files, such as "Announcement Summary" and "Residents Feedback Summary", are displayed clearly in a list format, each accompanied by a specific date and a "Download File" button for offline access and archiving.



The screenshot displays the 'Generate Report Form' within an administrative dashboard. The interface includes a sidebar on the left with navigation options: 'BARANGAY 133', 'Dashboard', 'Announcement Management', 'Residents Activity', 'Reports' (highlighted), 'Feedback/ Inquiries', and a 'Logout' button. The main content area is titled 'Summary Reports' and contains input fields for 'Start Date' and 'End Date' (both with date pickers), a 'Report Type' dropdown menu (currently showing '-- Select --'), and a 'File Type' label. A prominent blue 'Generate Report' button is located at the bottom right of the form. The footer of the dashboard shows the email 'barangay133@gmail.com' and the contact number '02XXX-03XXX'.

Figure 4.1.1 Generate Report Form

Figure 4.1.1 displays the system's Generate Report Form; the specific interface used for creating customized administrative summaries. It maintains the clean and consistent admin dashboard layout with the sidebar navigation. The key function of this tool is to allow officials to extract and export specific system data for analysis. The layout features input fields for "Start Date" and "End Date" to define the timeframe, and a "Report Type" dropdown menu offering categories such as "Announcement," "Feedback," "Activity History," and "Garbage Alert," supported by a prominent "Generate Report" button to process and compile the file.

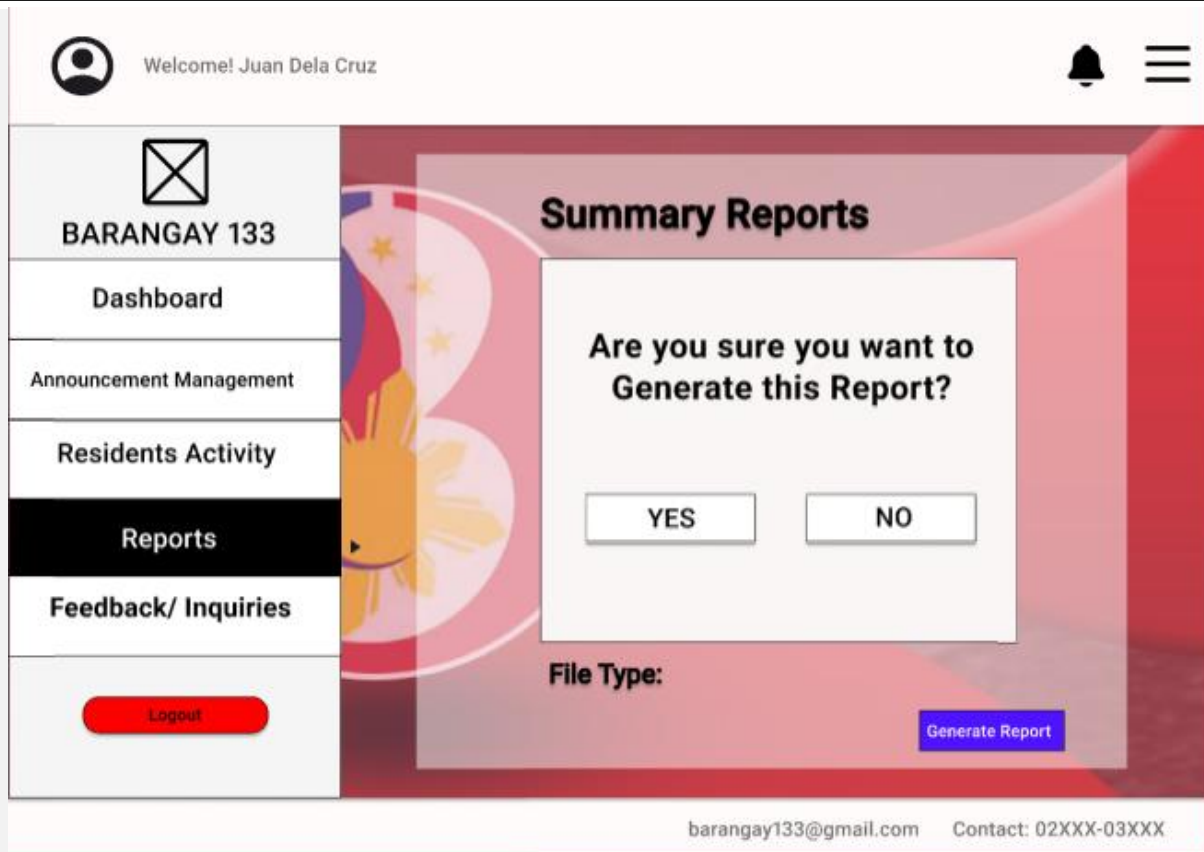


Figure 4.1.2 Generate Report Confirmation Dialog

Figure 4.1.2 This figure shows the Generate Report Confirmation Dialog within the Summary Reports module. The dialog appears when a user initiates the report generation process and serves as a confirmation step before proceeding. The message, *"Are you sure you want to Generate this Report?"*, prompts the user to verify their action to prevent unintended report generation. The dialog provides two response options, YES and NO, allowing the user to either continue with the report generation or cancel the process. A File Type selection option and a Generate Report button are available to define the output format and execute the report creation.

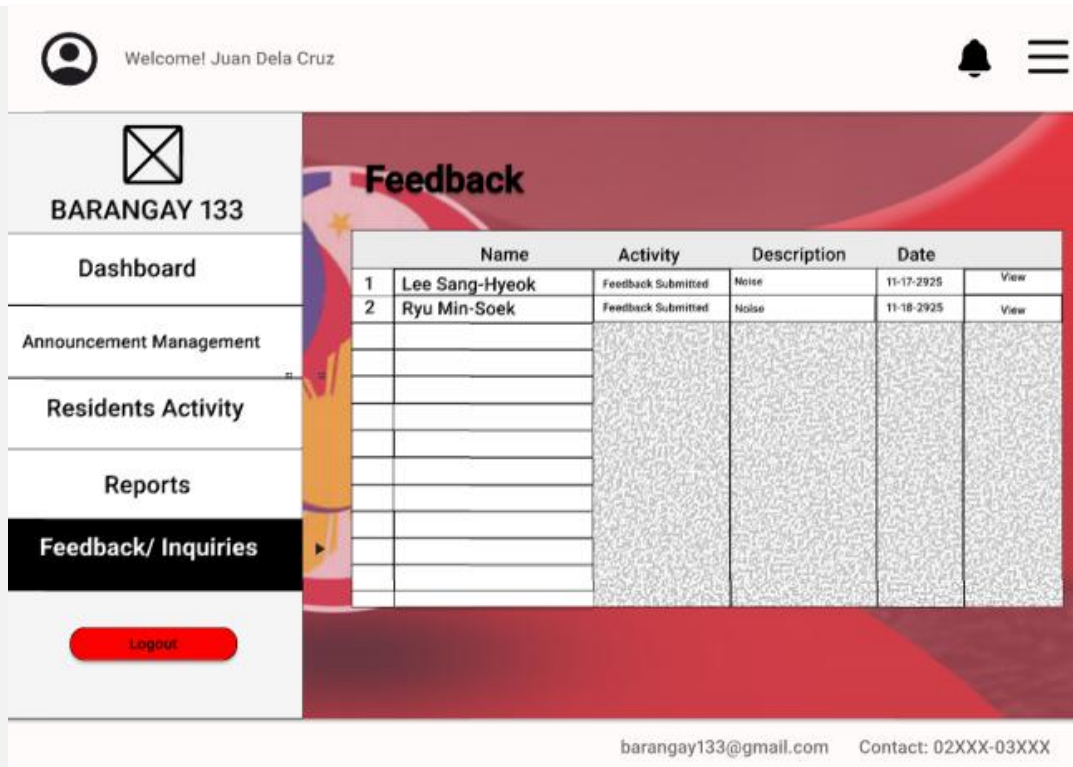


Figure 4.1.3 Feedback / Inquiries Interface

Figure 4.1.3 This figure presents the Feedback / Inquiries Interface, which allows authorized users to view and manage feedback submitted by residents through the system. The interface follows the standard system layout, with a left-side navigation panel highlighting the active Feedback / Inquiries module for clear user orientation. The main content area displays feedback records in a tabular format, including columns for Name, Activity, Description, and Date. Each entry reflects a submitted feedback item, such as noise-related concerns, providing administrators with organized and easily accessible information. A View option is available for each record, allowing users to review the full details of the submitted feedback.

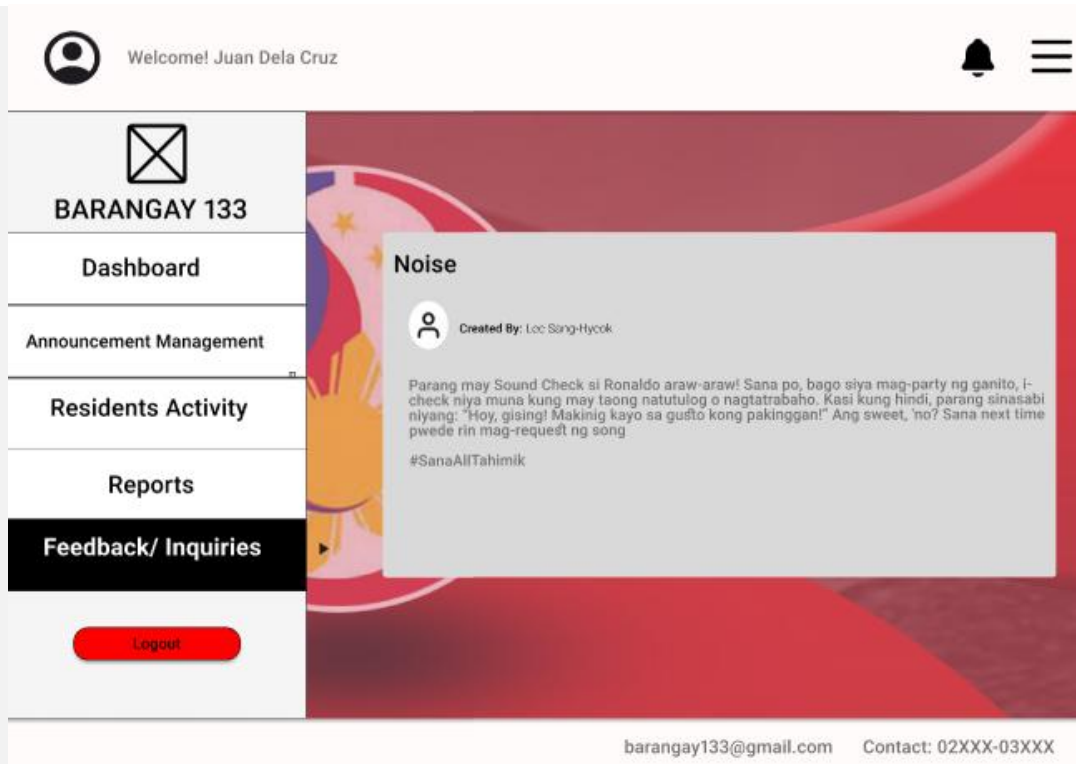


Figure 4.1.4 Feedback Details Page

Figure 4.1.4 displays the system's dedicated Feedback Details Page; the interface designed for reviewing specific resident concerns. It maintains the clean and consistent admin dashboard layout with the sidebar navigation clearly highlighting the active section. The key function of this view is to provide officials with the full context of a submitted report. The content area presents the specific issue, clearly identifying the category "Noise" and the source "Created By: Lee Sang-Hyeok", followed by a read-only text block containing the detailed message regarding the disturbance, ensuring officials have complete information for assessment.

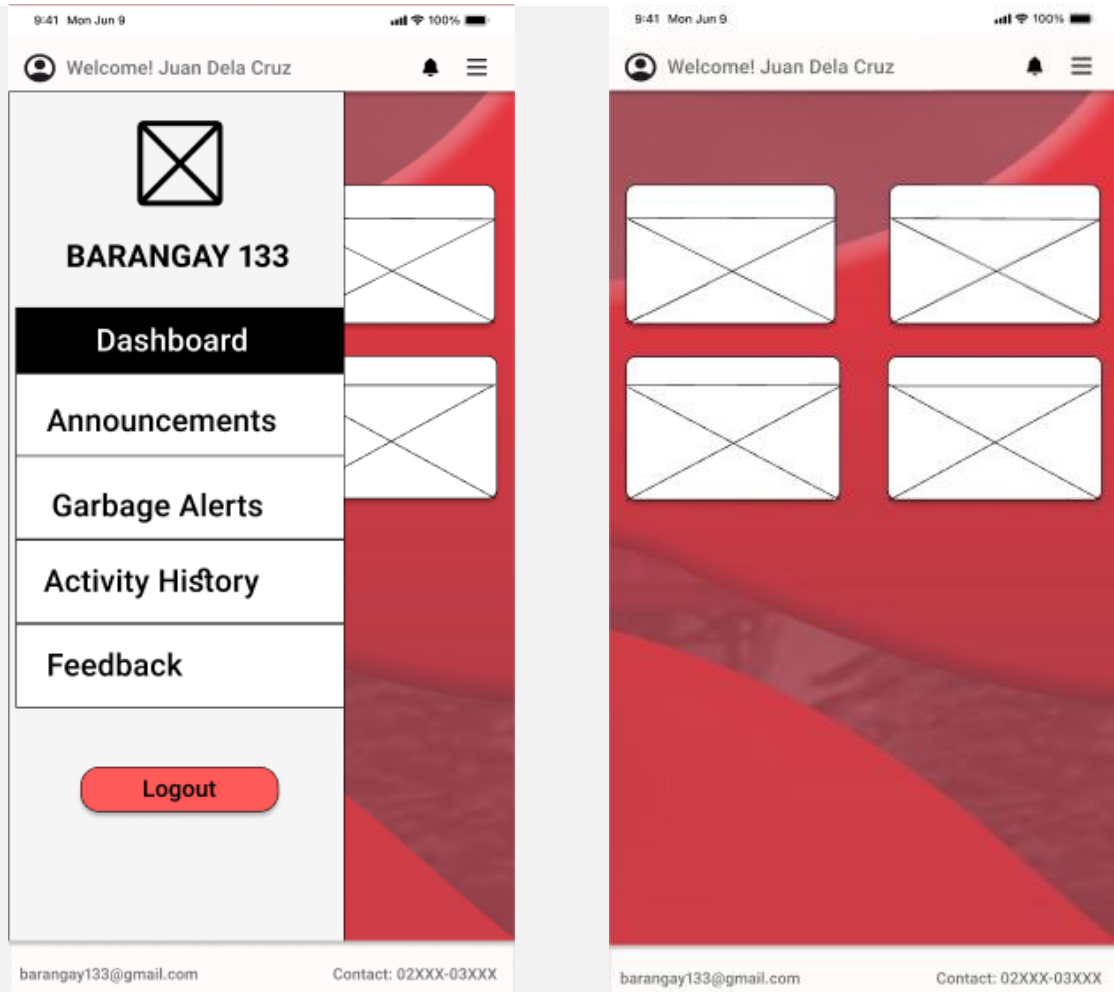


Figure 4.1.5 Resident Dashboard Interface

Figure 4.1.5 This figure illustrates the Mobile User Dashboard Interface of the Barangay 133 system, optimized for smartphone devices. The interface maintains consistency with the web-based version, displaying a personalized greeting, *"Welcome! Juan Dela Cruz,"* along with notification and menu icons at the top of the screen. On the left side, a collapsible navigation menu provides access to key modules, including Dashboard, Announcements, Garbage Alerts, Activity History, and Feedback, as well as a Logout option at the bottom. This menu allows users to easily navigate between features within the mobile application. The main content area presents information in a card-based layout, enabling users to quickly view announcements, alerts, and updates in a clear and organized manner. The responsive design ensures that content is displayed effectively across different screen sizes, supporting accessibility and ease of use.

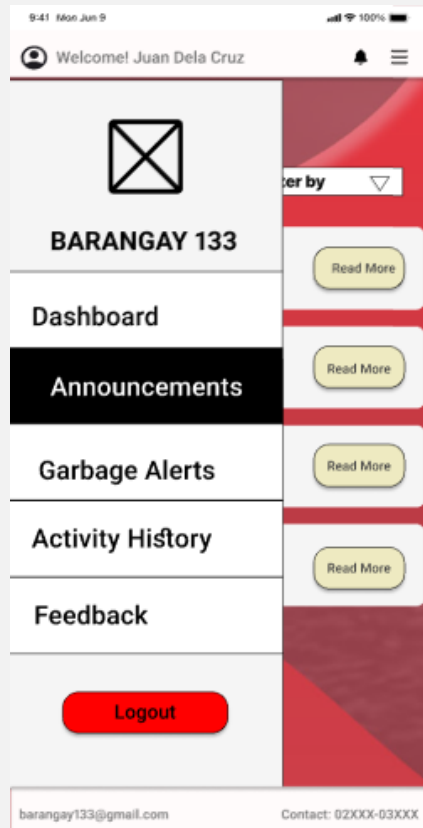


Figure 4.1.6 Mobile Announcements Interface

Figure 4.1.6 This figure presents the Mobile Announcements Interface of the Barangay 133 system. The screen is optimized for mobile devices and maintains a consistent layout with the desktop version. A personalized greeting, *"Welcome! Juan Dela Cruz,"* is displayed at the top, along with notification and menu icons for easy access. A vertical navigation menu is shown on the left side, providing access to Dashboard, Announcements, Garbage Alerts, Activity History, and Feedback, with the Announcements section highlighted to indicate the active module. The main content area displays announcement items in a stacked, card-style format, each accompanied by a "Read More" button to allow users to view full details.

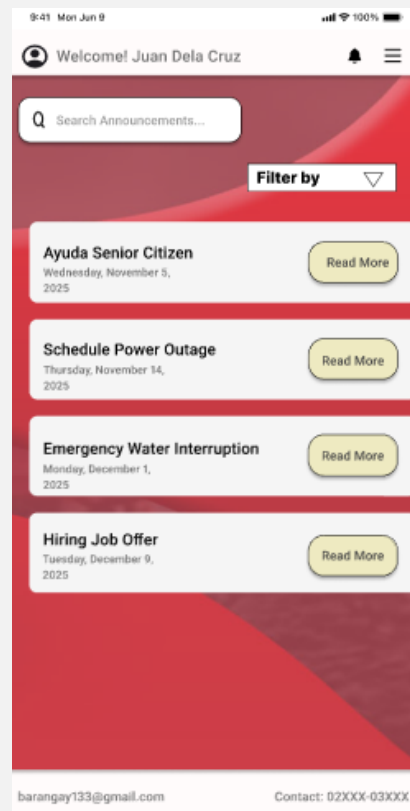


Figure 4.1.7 Mobile Announcements Interface Continuation

Figure 4.1.7 presents the Mobile Announcements List Interface of the Barangay 133 system. This screen displays a searchable and filterable list of barangay announcements optimized for mobile devices. A Search Announcements field is provided at the top to allow users to quickly locate specific announcements, while a Filter by option enables sorting based on selected criteria. The main content area shows announcement items in a vertical card-based layout, including the announcement title, date, and a “Read More” button for viewing complete details. Sample announcements displayed include *Ayuda Senior Citizen*, *Schedule Power Outage*, *Emergency Water Interruption*, and *Hiring Job Offer*. This interface ensures convenient access to important barangay updates and supports efficient information browsing for mobile users.



Figure 4.1.8 Mobile Announcements Interface Continuation

Figure 4.1.8 presents the Mobile Announcement Details Interface, which is displayed when a user selects the “Read More” option from the mobile announcements list. This screen provides a detailed view of a selected announcement, such as “Ayuda Senior Citizen Details”, including the announcement title, date, and full descriptive content. The interface organizes information into clearly defined sections, including the Announcement Summary, Schedule and Distribution Details, and Requirements and Important Reminders, allowing users to easily understand eligibility, location, and necessary documents. A back navigation button is provided at the top of the screen to allow users to return to the announcements list.

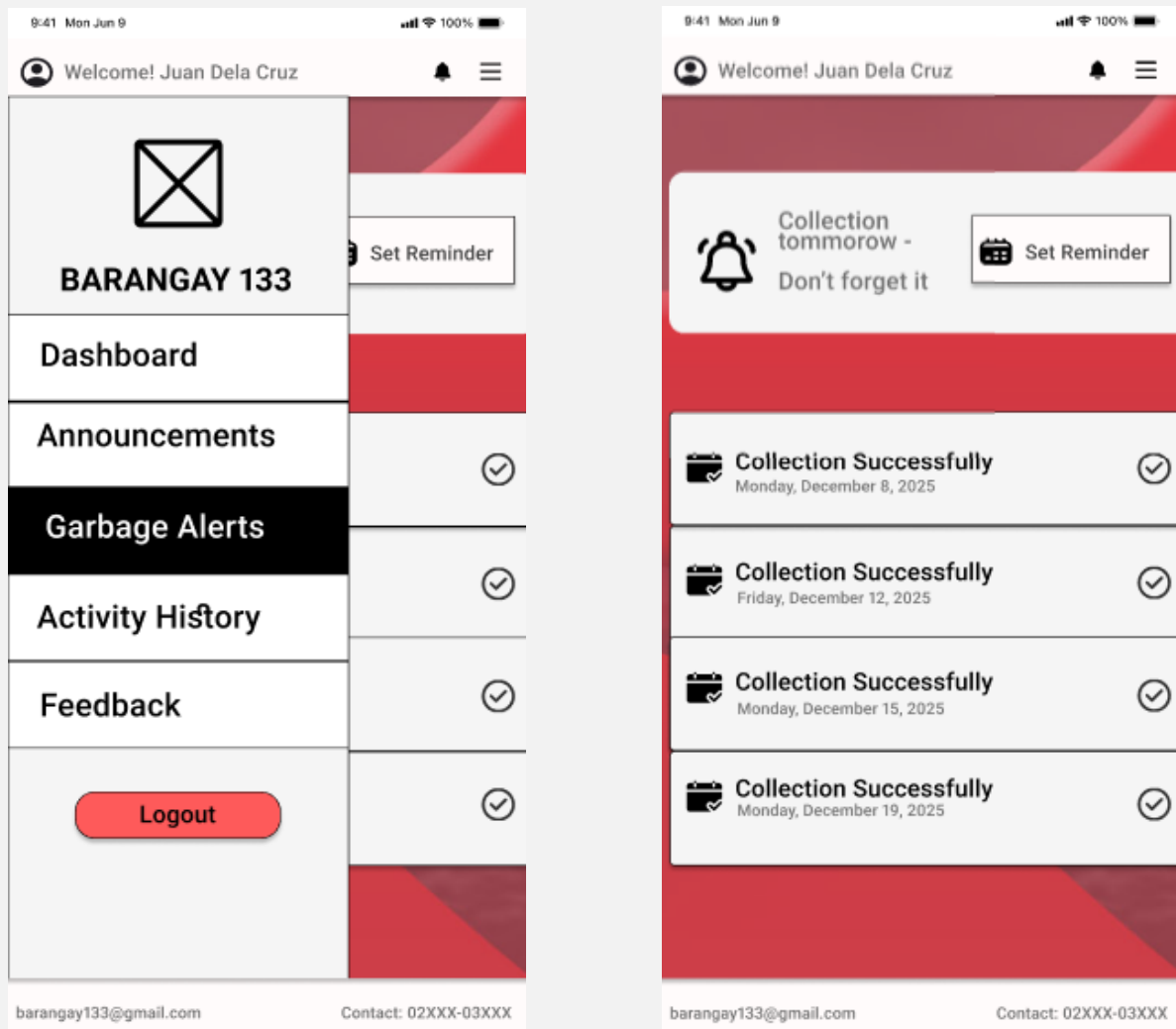


Figure 4.1.9 Garbage Alert Interface

Figure 4.1.9 presents the Garbage Alert Interface, a mobile-optimized module designed to facilitate real-time sanitation updates for residents. The interface is divided into two primary functional zones: a predictive alert banner and a historical service log. The upper section features a proactive notification card (e.g., "Collection tomorrow") equipped with a "Set Reminder" utility, enabling users to schedule personal alerts to ensure timely waste disposal. The lower section maintains a chronological record of completed service rounds, labeled "Collection Successfully" and accompanied by specific timestamps (e.g., "Monday, December 8, 2025"). This design not only assists residents in adhering to disposal schedules but also provides a transparent, verifiable history of municipal service performance.

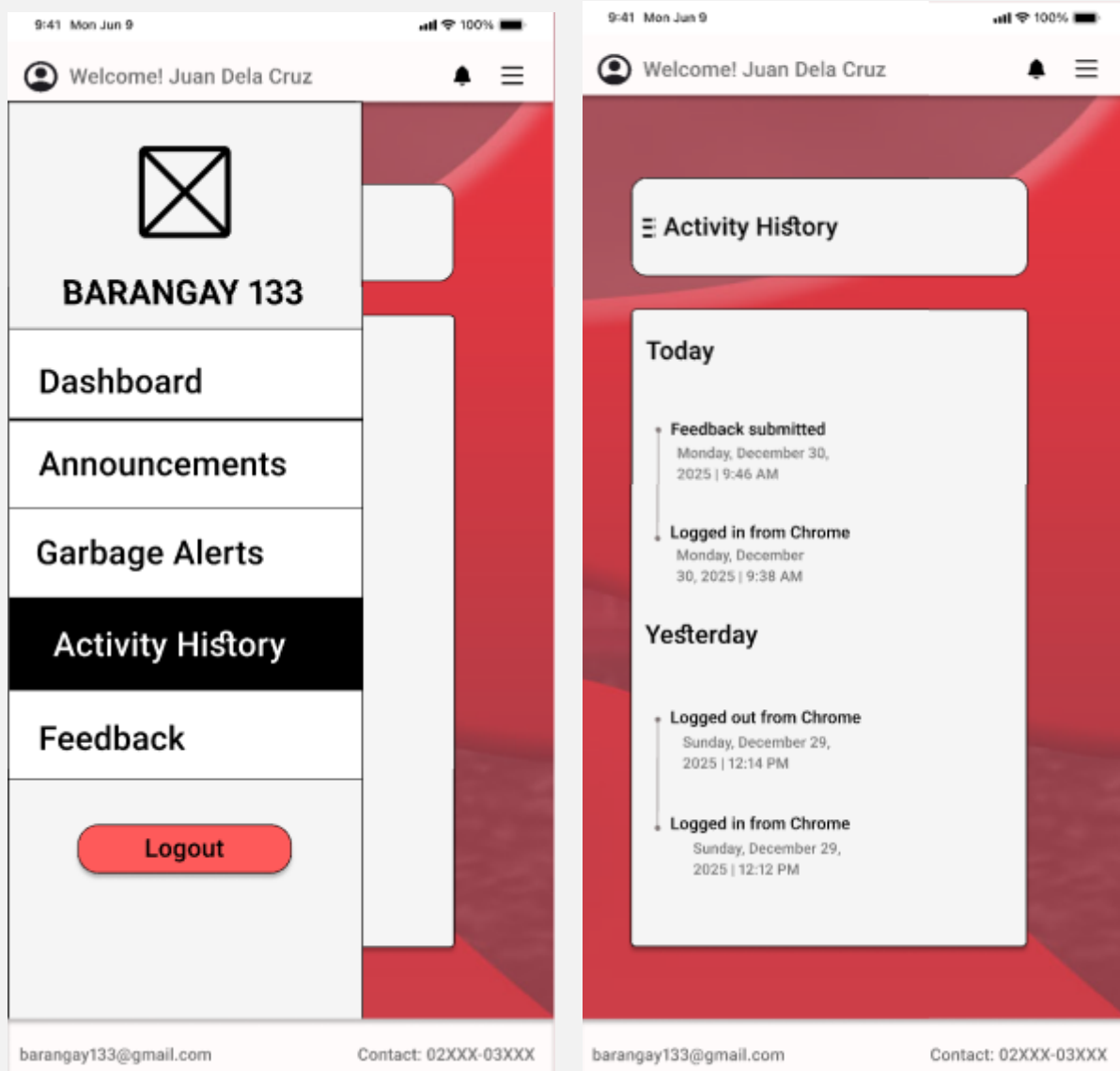


Figure 4.2.0 Activity History Interface

Figure 4.2.0 depicts the Activity History Interface, a user-centric auditing module designed to enhance account transparency and personal security. The interface employs a vertical timeline layout to present a chronological log of user interactions, visually grouped by temporal segments such as "Today" and "Yesterday." Each entry records specific system events—ranging from authentication activities like "Logged in from Chrome" to operational actions such as "Feedback submitted"—accompanied by precise timestamps. This design empowers residents to monitor their account usage patterns in real-time, allowing them to verify their own submissions and quickly identify any unrecognized access or discrepancies in their activity feed.

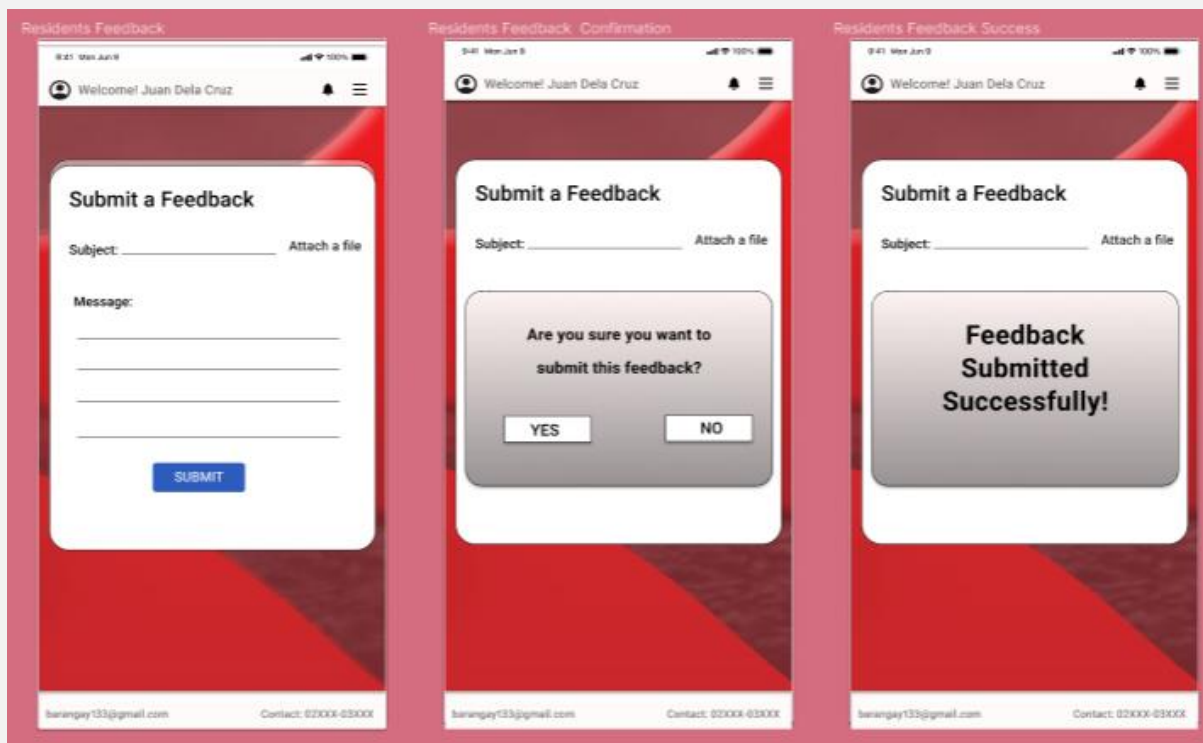


Figure 4.2.1 Feedback Interface

Figure 4.2.1 illustrates the Feedback Interface, a sequential interaction workflow designed to facilitate structured communication between residents and barangay officials. The figure delineates the three-stage process for lodging complaints or suggestions, ensuring data accuracy and user confidence. The workflow initiates with the Composition Screen (Left), where residents articulate their concerns via the Subject and Message input fields; crucially, this form includes an "Attach a file" utility, empowering users to upload digital evidence (such as photographs of uncollected garbage) to substantiate their reports. Upon actuating the "SUBMIT" command, the system transitions to the Confirmation Screen (Center), which triggers a modal dialog querying, "Are you sure you want to submit this feedback?"—a safeguard designed to prevent accidental transmissions. Finally, the process concludes with the Completion Screen (Right), which displays a prominent "Feedback Submitted Successfully!" notification, providing the user with immediate visual verification that their report has been securely logged into the central database for administrative review.

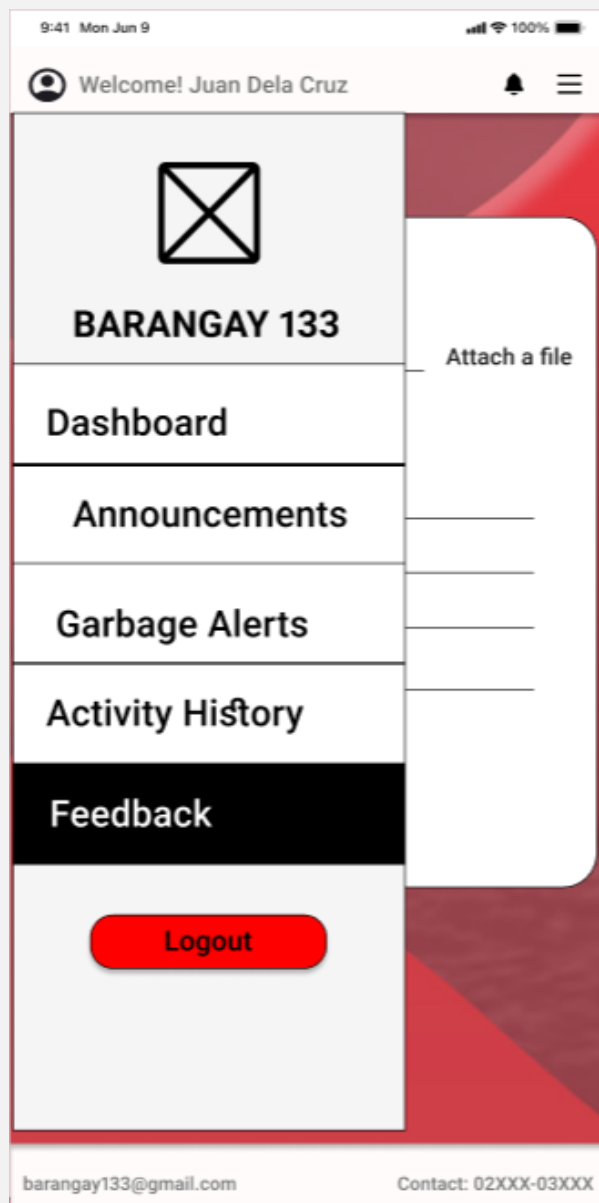


Figure 4.2.2 Feedback Side Menu

Figure 4.2.2 presents the Feedback Side Menu, the primary navigational framework designed to ensure seamless mobility within the resident's mobile interface. The figure captures the menu's active state, where the "Feedback" module is visually highlighted in black to indicate the user's current location within the application hierarchy. This collapsible navigation drawer aggregates all critical system functions—including "Dashboard," "Announcements," "Garbage Alerts," and "Activity History"—into a unified, accessible list, facilitating rapid transitions between different operational modules. Furthermore, the interface reinforces session security and personalization by displaying the authenticated user's identity ("Welcome! Juan Dela